High Exponents May Not Suffice to Patch AIM

On Attacks, Weak Parameters, and Patches for AIM2

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Abstract. The growth of advanced cryptographic applications has driven the development of arithmetization-oriented (AO) ciphers over large finite fields, which are designed to minimize multiplicative complexity. However, this design advantage of AO ciphers could also serve as an attack vector. For instance, the AIM one-way function in the post-quantum signature AIMer proposed at CCS 2023 has been broken by several works soon after its publication. The designers then promptly developed secure patches and proposed an enhanced version, AIM2, which was updated to the latest version of AIMer that has been selected as one of the winners of the Korean PQC Competition in early 2025.

In this paper, we focus on the algebraic security of AIM2 over $\mathbb{F}_{2^n}$. First, we introduce a resultant-minimized model that reduces eliminations by using a non-k based substitution strategy and linearized-polynomial decomposition, achieving an attack time complexity of $2^{188.76}$ ($2^{195.05}$) primitive calls of AIM2-III when $\omega = 2$ ($\omega = 2.373$), indicating that the designers have been over-optimistic in the evaluation of their security margin; Second, we propose a subfield reduction technique for the case that exponents approach subfield extension sizes, equation degrees collapse sharply, *e.g.,* the exponent $e_2 = 141 \mapsto 13$ in AIM2-V when considering the subfield $\mathbb{F}_{2^{128}}$. This can lower the algebraic attack complexity to $2^{295.97}$ primitive calls at $\omega = 2$, which improves upon designers' estimated bound of Gröbner basis attack by about 2^{100}. Besides, based on our attack methods, we have identified some weak parameter choices, which could provide concrete design guidance for AIM2 construction, especially for the exponent of its Mersenne S-box. Finally, to address the potential vulnerabilities, we further propose AIM2-patch with a simple secure patch on AIM2, which can prevent key elimination, neutralize linearized-polynomial decomposition, and raise algebraic attack complexity, while incurring negligible overheads in AIMer scheme.

Keywords: AIMer· AIM2 · Resultant Attack · Resultant-minimized Modeling · Subfield Reduction · Linearized Polynomial Decomposition

1 Introduction

In recent years, privacy-preserving protocols such as multi-party computation (MPC), fully homomorphic encryption (FHE), and zero-knowledge proofs (ZK) have attracted significant attention. The efficiency of these protocols is largely determined by the number and type of nonlinear operations they require. This new cost metric has led to the rise of *arithmetic-oriented* (AO) symmetric primitives, which optimize efficiency by minimizing nonlinear operations to improve circuit efficiency. Since LowMC [3] proposed at EUROCRYPT 2015, many such designs have been proposed, including Rescue [4], Poseidon [19], Griffin [16], Reinforced Concrete [17], and Monolith [18], targeting prime fields; Jarvis [5], Vision [4], Rain [14], and AIM [23], over binary extension fields; as well as MiMC [1], Ciminion [13], and Anemoi [10], which operate over both domains.

The development of post-quantum cryptography (PQC) further amplifies the importance of AO primitives. For instance, the MPC-in-the-Head (MPCitH) framework of Ishai *et al.* [20] provides a post-quantum signature construction based on non-interactive ZK proofs, where the underlying cost is driven by multiplication complexity. LowMC has already been deployed in the PICNIC signature scheme [11], a third-round candidate in the NIST PQC competition. This approach was later extended to AIMer [23] at CCS 2023, which is an MPCitH-based signature scheme selected for NIST PQC Round 1 Additional Signatures[5] and the winner of the KpqC competition[6]. In the AIMer signature scheme, a cipher called AIM was designed as its underlying one-way function, by which a single known pair $(iv, ct = \mathsf{AIM}(k, iv))$ is made public as the verification key, while the secret key k serves as the signing key. Thus, if an adversary can recover k from (iv, ct), they can trivially forge signatures by simulating the honest prover, and AIM was explicitly designed to resist this attack setting.

However, the algebraic structure of AO primitives makes them more vulnerable to algebraic attacks. The hardness of solving multivariate polynomial systems derived from their algebraic descriptions is thus typically considered as the main metric to evaluate their security. Recent advances have markedly improved such attacks, including Gröbner basis attacks [2,8,26,7], resultant-based elimination methods from ASIACRYPT 2024 [29], and refined resultant attacks at CRYPTO 2025 [6]. These results have advanced the cryptanalysis of AO designs, showing that their design advantage, *i.e.*, algebraic properties, can also serve as an attack vector.

Related Works. The simple and clean construction of AIM quickly drew cryptanalytic interest. Zhang *et al.* first proposed a guess-and-determine attack on full AIM-III (the variant targeting for 192 bits of security) at ASIACRYPT 2023 [30].

[5] https://csrc.nist.gov/Projects/pqc-dig-sig/round-1-additional-signatures
[6] https://kpqc.or.kr/

The main technique in [30] is to linearize S-boxes by guessing the power of the input over $\mathbb{F}_2^n$ and reduce the attack to solving a system of linear equations. In a parallel work at FSE 2024, by employing the fast exhaustive search method [9] to achieve better time-memory trade-offs, Liu *et al.* [27] significantly improved the time complexity for full AIM. All the aforementioned algebraic attacks on AIM have been conducted over the binary field $\mathbb{F}_2$.

Kim *et al.* [22] then prompted to propose a patched version of AIM2 to mitigate the previous attacks. Besides modifying the design, they also carried out a thorough algebraic cryptanalysis using Gröbner basis attacks. Such bounds are often adopted by designers as reference security evaluations, *i.e.*, the higher the derived attack complexity, the greater confidence in the claimed security of the target cipher. More recently, Yang *et al.* [28] advanced this direction by introducing a polynomial decomposition method and mounting the first successful algebraic attack on full AIM over the extension field $\mathbb{F}_{2^n}$.

Our Contributions. In this work, we focus on AIM2, providing more accurate security evaluations against state-of-the-art algebraic attacks and proposing refinements that exploit its inherent algebraic properties. Based on these analyses, we identify weak parameter choices and give concrete design guidance, including a patch that strengthens AIM2 with negligible efficiency loss.

As can be seen in AIM2 [22], the designers carefully modified exponents and constants to prevent previously successful algebraic attacks over $\mathbb{F}_2$, which also prevents the attack model of Yang *et al.* [28]. It can be noted that the designers' security evaluations of AIM2 mainly relies on constructing polynomial systems over $\mathbb{F}_2$ and analyzing them with Gröbner basis attacks, a common baseline for designers, where higher derived bounds give more confidence in security. However, such approaches overlook specific algebraic structures of AIM2 and risk optimistic estimates. Besides, extending Yang *et al.*'s framework to AIM2 also suffers from degree blow-up during successive resultants, leading to worse complexity than the designers' own evaluations of Gröbner basis attacks.

To overcome these issues, we introduce new algebraic tools tailored for AIM2, namely a resultant-minimized modeling that reduces eliminations and exploits low-degree decompositions, and subfield reduction method that leverages near-subfield exponents to cut equation degrees dramatically. Collectively, these methods yield smaller, lower-degree systems and improved attack time complexity bounds, showing that the designers' evaluations of AIM2 may have been optimistic, as summarized in Table 1.[7] To be more specific, for AIM2-III, we have

[7] In this paper we evaluate attack complexities in the number of field operations, for a fair comparison with [22], which will be converted into the number of encryptions. The conversions are referenced as below

- A field multiplication over $\mathbb{F}_{2^n}$ is equivalent to $n \log n$ bit operations [12],
- A field addition over $\mathbb{F}_{2^n}$ is equivalent to n bitwise XOR operations,
- One call of primitives over $\mathbb{F}_{2^n}$ is equivalent to n^3 bit operations.

For simplicity, we evaluate one field operation by $n \log n$ bit operations.

achieved attack time complexities close to the designers' claimed security when $\omega = 2$ or 2.373, thus questioning its security margin; for AIM2-V, we have proposed a subfield reduction method combined with resultant-minimized modeling, which lowers the attack complexity by around 2^{100} when compared to designers' algebraic attacks [22]. In the following, we introduce our main contributions to further explain these results.

Table 1: Time complexities of AIM2 and AIM2-patch against algebraic attacks, which are given in the number of equivalent primitive calls.

Primitives	Techniques	T $\omega = 2$	T $\omega = 2.373$	T $\omega = 2.8$	Ref.
AIM2-III	Gröbner basis attack†	239.65	288.2	344.25	[22]
	linearized polynomial decomposition	216.76	234.29	254.35	[28]‡
	resultant-minimized modeling	**188.76**	**195.05**	**202.36**	Sec. 3.1
AIM2-V	Gröbner basis attack	390.4	466.4	549.5	[22]
	linearized polynomial decomposition	449.43	502.02	562.23	[28]
	subfield-based reduction	**295.97**	**305.81**	**317.24**	Sec. 3.2
AIM2-patch-III*	resultant	284.34	301.87	321.94	Sec. 4.3
AIM2-patch-V	subfield-based reduction	336.27	341.49	349.91	Sec. 4.3
AIM2-patch-I	resultant	316.57	342.74	381.59	Sec. 4.3

† We recalculate the complexity of [22] in the number of equivalent calls of the primitives, *e.g.*, one encryption of AIM2 takes about $\frac{n^2}{\log n}$ field-operations.

‡ We evaluated the security bounds of AIM2 using the attack models in [28].

* The claimed security of AIM2-patch over $\mathbb{F}_2$ is consistent with that of AIM2.

Resultant-minimized Modeling. Our attack model follows the algebraic meet-in-the-middle (MITM) framework of prior works [28], but we refine it with a dedicated substitution strategy and a resultant-minimized structure. Specifically, we reformulate AIM2 equations over $\mathbb{F}_{2^n}$ to reduce the number of variables and limit the resultant computations required after eliminating the input k, *i.e.*, we adopt a non-k-based substitution strategy. We further exploit the decomposition of linearized polynomials to keep degrees small, and carefully select v_2 as the base variable for substitution, which will be the optimal choice in terms of the number of resultant computations. Applying this strategy to AIM2-III, we obtain a complexity of $2^{188.76}$ primitive calls when $\omega = 2$. This shows that, under standard assumptions from the designers' perspective, the claimed security margins of AIM2-III will not be enough.

Subfield Reduction Technique. By leveraging an algebraic property of finite fields $\mathbb{F}_{2^n}$ and their subfields: for any $\mathbb{F}_{2^{n/R}} \subset \mathbb{F}_{2^n}$, every nonzero polynomial

$f(x) \in \mathbb{F}_{2^{n/R}}$ satisfies $\deg(f) \leq 2^{n/R} - 1$ due to the relation $x^{2^{n/R}} = x$, we introduce the subfield reduction technique for analyzing AIM2 instances, where some exponents e_i are close to the subfield extension size. Applying this method to AIM2-V targeting 256-bit security, we reduce equation degrees from 2^{141} to 2^{13} by working over the subfield $\mathbb{F}_{2^{128}}$. To achieve this, we construct linearized polynomials with algebraic structures that preserve low-degree forms over $\mathbb{F}_{2^{128}}$, and employ a fast substitution strategy (inspired by [28]). Once two bivariate equations are derived via resultant elimination, we guess one variable and solve the other using the half-gcd algorithm [15].

Combining this method with our resultant-minimized modeling, we have achieved a complexity reduction of about 2^{100} compared to designers' given attack bounds [22], showing that the original security estimates of AIM2-V against algebraic attacks were somewhat optimistic.

Weak Parameters and Secure Patches for AIM2. Our attacks mentioned above highlight two potential vulnerabilities for AIM2 cipher: (i) the resultant-minimized modeling and non-k-based substitution show that some exponents in AIM2 remain small to withstand algebraic attacks, and (ii) the subfield reduction method reveals that exponents close to subfield extension sizes can significantly lower the security.

From these observations, we have identified concrete weak parameter choices and provide design guidance for selecting secure ones. To further improve the security of the construction, we propose a lightweight AIM2-patch that incorporates an additional linear layer L_k immediately after the key addition. This ensures that k variable cannot be easily eliminated from the system of equations, thereby blocking linearized polynomial decomposition and greatly increasing algebraic attack complexity. Importantly, AIM2-patch preserves all original parameters and irreducible polynomials, while incurring only negligible performance overhead in AIMer. Thus, it offers a simple yet effective means of mitigating the identified vulnerabilities without sacrificing efficiency.

Organization. The rest of the paper is organized as follows. In Section 2, some background knowledge will be introduced. Then, in Section 3, we introduce the resultant-minimized modeling and the subfield reduction technique, which could provide more accurate security evaluations of AIM2 against algebraic attacks. Later, in Section 4, some weak parameters and a simple secure patch of AIM2 are provided, both of which could serve as design guidance for AIM2 construction. Finally, we conclude the paper in Section 5.

2 Preliminaries

2.1 Notation

Let $\mathbb{F}_q = \mathbb{F}_{p^n}$ denote a finite field of prime characteristic p, where $q = p^n$ and $n \in \mathbb{N}$. For AIM2 defined over $\mathbb{F}_{2^n}$, we use n, λ, and ℓ to represent the block size in bits, the security parameter, and the number of parallel branches, respectively.

Consider the multivariate polynomial ring $R = \mathbb{F}_{2^n}[x_1, \ldots, x_n]$, each polynomial $P(x_1, \ldots, x_n) \in R$ can be written as

$$P(x_1, \ldots, x_n) = \sum_{\alpha_1=0}^{2^n-1} \cdots \sum_{\alpha_n=0}^{2^n-1} u_{\alpha_1 \cdots \alpha_n} \, x_1^{\alpha_1} \cdots x_n^{\alpha_n},$$

where $u_{\alpha_1 \cdots \alpha_n} \in \mathbb{F}_{2^n}$ are the coefficients, and $x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ is called a term. The total degree of P is defined as

$$\deg(P) = \max \left\{ \sum_{i=1}^{n} \alpha_i \;\middle|\; u_{\alpha_1 \cdots \alpha_n} \neq 0 \right\}.$$

The degree of P with respect to the variable x_i is

$$\deg_{x_i}(P) = \max \left\{ \alpha_i \mid u_{\alpha_1 \cdots \alpha_n} \neq 0 \right\}.$$

For any polynomial P, we write P' to represent the version after the substitution of P, not the derivative. In this paper, unless otherwise specified, all logarithms are taken in base 2.

2.2 Linearized Polynomial

Definition 1 (Linearized Polynomial). *A* linearized polynomial $L(x)$ *over* $\mathbb{F}_{p^n}$ *is a polynomial of the form*

$$L(x) = \sum_{i=0}^{n-1} \alpha_i x^{p^i}, \qquad \alpha_i \in \mathbb{F}_{p^n}.$$

It is immediate that if $L_1(x)$ and $L_2(x)$ are linearized polynomials, then their composition $L_3(x) = L_1 \circ L_2(x)$ is also a linearized polynomial. For all $a, b \in \mathbb{F}_{p^n}$ and $c \in \mathbb{F}_p$, the following properties hold:

$$L(a+b) = L(a) + L(b), \qquad c \cdot L(a) = L(c \cdot a).$$

Moreover, due to the fact that $\mathbb{F}_{p^n}$ and $\mathbb{F}_p^n$ are isomorphic, any $x \in \mathbb{F}_{p^n}$ can be represented as a vector $\hat{x} \in \mathbb{F}_p^n$. In particular, there exists a matrix $M \in \mathbb{F}_p^{n \times n}$ such that $L(x) = M\hat{x}$. Besides, note that not every linearized polynomial is a permutation polynomial. For $L(x)$ to admit an inverse $L^{-1}(x)$, it must satisfy specific invertible conditions, which will be given in the following theorem.

Theorem 1. *A linearized polynomial $L(x) = \sum_{i=0}^{n-1} \alpha_i x^{p^i}$ is a permutation polynomial if and only if its Dickson matrix $D_L \in \mathbb{F}_{p^n}^{n \times n}$ is invertible, where*

$$D_L = \begin{pmatrix} \alpha_0 & \alpha_1 & \cdots & \alpha_{n-1} \\ \alpha_{n-1}^p & \alpha_0^p & \cdots & \alpha_{n-2}^p \\ \vdots & \vdots & \vdots & \vdots \\ \alpha_1^{p^{n-1}} & \alpha_2^{p^{n-1}} & \cdots & \alpha_0^{p^{n-1}} \end{pmatrix}.$$

Moreover, when D_L is invertible, the inverse linearized polynomial of $L(x)$ can be derived by

$$L^{-1}(x) = \frac{1}{\det(D_L)} \sum_{i=0}^{n-1} \bar{\alpha}_i \, x^{p^i},$$

where $\bar{\alpha}_i$ denote the $(i,0)$-cofactor of D_L.

Remark. The decomposition of linearized polynomials is crucial for resultant-based algebraic attacks (later introduced in Section 2.3), as it helps to reduce equation degrees and facilitates elimination techniques, which is detailed below.

Proposition 1 ([28]). *Given a linearized polynomial $L(x) = \sum_{i=0}^{n-1} x^{2^i}$ over $\mathbb{F}_{2^n}$, if there exists a the low-degree permutation polynomial $L'(x)$ such that $L''(x) = L' \circ L(x)$ also has a low degree, then $L(x)$ is said to admit a decomposition of the form $L(x) = L'^{-1} \circ L''(x)$.*

2.3 Resultant-based Solving Method

Algebraic attacks on symmetric-key primitives are often reduced to solving a system of m multivariate polynomial equations in n variables over $\mathbb{F}_q$, namely

$$\begin{cases} f_1(x_1, x_2, \ldots, x_n) = 0, \\ f_2(x_1, x_2, \ldots, x_n) = 0, \\ \quad \vdots \\ f_m(x_1, x_2, \ldots, x_n) = 0. \end{cases} \tag{1}$$

The ideal generated by $f_1, \ldots, f_m$ is defined as

$$I := \langle f_1, \ldots, f_m \rangle = \left\{ \sum_{i=1}^{m} f_i g_i \;\middle|\; g_i \in \mathbb{F}_q[x_1, \ldots, x_n] \right\}.$$

In this subsection, we revisit a state-of-the-art method for solving systems of multivariate equations via variable elimination, which will be instrumental in our later attacks on AIM2. Specifically, if we regard $f_i(x_1, x_2, \ldots, x_n)$ as a univariate polynomial in the indeterminate x_n, it can be rewritten in the form $f_i(\boldsymbol{y}, x_n)$ and $\boldsymbol{y} = (x_1, \ldots, x_{n-1})$. Moreover, $f_i(\boldsymbol{y}, x_n)$ admits the expression $f_i(\boldsymbol{y}, x_n) = \sum_{k=0}^{\delta_i} a_{i,k}(\boldsymbol{y}) \, x_n^k$, where $a_{i,k} \in \mathbb{F}_q[\boldsymbol{y}]$ and $\delta_i = \deg_{x_n}(f_i(\boldsymbol{y}, x_n))$.

Definition 2 (Resultant of Two Polynomials). *Given two polynomials $f_1(\boldsymbol{y}, x_n) = \sum_{k=0}^{\delta_1} a_{1,k} x_n^k$ and $f_2(\boldsymbol{y}, x_n) = \sum_{k=0}^{\delta_2} a_{2,k} x_n^k$, where $a_{1,k}, a_{2,k} \in \mathbb{F}_q[\boldsymbol{y}]$,* *the* Sylvester matrix *of* f_1, f_2 *with respect to* x_n *is the* $(\delta_1+\delta_2)\times(\delta_1+\delta_2)$ *matrix*

$$\mathrm{Syl}(f_1, f_2, x_n) = \begin{bmatrix} a_{1,\delta_1} & \cdots & a_{1,1} & a_{1,0} & & & & \\ & a_{1,\delta_1} & \cdots & a_{1,1} & a_{1,0} & & & \\ & & \ddots & \ddots & \ddots & \ddots & & \\ & & & a_{1,\delta_1} & \cdots & a_{1,1} & a_{1,0} & \\ a_{2,\delta_2} & \cdots & a_{2,1} & a_{2,0} & & & & \\ & a_{2,\delta_2} & \cdots & a_{2,1} & a_{2,0} & & & \\ & & \ddots & \ddots & \ddots & \ddots & & \\ & & & a_{2,\delta_2} & \cdots & a_{2,1} & a_{2,0} & \end{bmatrix},$$

where the first δ_2 *rows consist of shifted coefficient vectors of* f_1*, and the last* δ_1 *rows consist of shifted coefficient vectors of* f_2*. The* resultant *of* f_1 *and* f_2 *with respect to* x_n *is defined as* $\mathrm{Res}(f_1, f_2, x_n) = \det\big(\mathrm{Syl}(f_1, f_2, x_n)\big)$.

Under the assumption of algebraic independence, the resultant yields a new polynomial $h(\boldsymbol{y})$ such that $h(\boldsymbol{y}) = 0$ if and only if $(\boldsymbol{y}, x_n)$ is a common root of both f_1 and f_2. This enables variable elimination between two polynomials without loss of information. Specifically, we compute

$$\mathrm{Res}(f_1, f_2, x_n),\ \mathrm{Res}(f_1, f_3, x_n),\ \ldots,\ \mathrm{Res}(f_1, f_m, x_n),$$

which eliminates x_n while preserving all solutions in variables $x_1, \ldots, x_{n-1}$ from Equation (1). By iteratively eliminating, the system can eventually be reduced to a univariate polynomial when $m = n$. Once the univariate polynomial is obtained, it can be solved directly, and the process is iterated to recover all common solutions.

Two key components in analyzing the complexity of the above resultant-based solving are the cost of multiplying two multivariate polynomials, and the cost of solving a univariate polynomial, which will be briefly introduced as below.

Fast Multivariate Polynomial Multiplication. The most efficient known algorithm for multiplying two univariate polynomials of degree d over $\mathbb{F}_q$ is based on the Fast Fourier Transform (FFT), achieving quasi-linear complexity

$$\mathcal{M}(d) = \mathcal{O}\big(d \log d \log\log d\big) \tag{2}$$

field operations. Given two polynomials $f, g \in \mathbb{F}_q[x_1, \ldots, x_n]$, where $\deg_{x_i}(f) = \alpha_i$ and $\deg_{x_i}(g) = \beta_i$ for $1 \le i \le n$. By applying the univariate multiplication algorithm together with the Kronecker substitution (see §3.4 of [21]), the product $f \cdot g$ can be computed in field operations as

$$\mathcal{M}\left(\prod_{i=1}^{n}(\alpha_i + \beta_i + 1)\right). \tag{3}$$

Root-Finding for Univariate Polynomials. Yang *et al.* [29] shows that finding the roots of a univariate polynomial $R(x) \in \mathbb{F}_q$ of degree d will take

$$\mathcal{U}(d) = \mathcal{O}\big(d \log d \,(\log d + \log q \log\log d)\big) \quad (4)$$

field operations. This complexity is based on the fast *half-gcd* algorithm [15], which itself runs in $\mathcal{O}(d(\log d)^2)$ field operations.

2.4 AIM2 One-way Function: An Enhanced Version of AIM

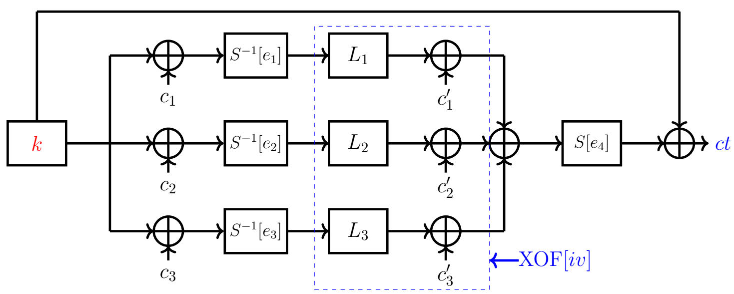

Fig. 1: AIM2-V cipher with $\ell = 3$ branches. The input k is the secret key of the signature scheme, and (iv, ct) is the corresponding public key.

As the AIM cipher has been fully broken by [27,30], its designers proposed secure patches and introduced an enhanced version, AIM2 [22], at the 5th PQC Standardization Conference. The parameters of AIM2 are flexible, each user may independently choose parameters and instantiate the cipher accordingly. Given a secret key k, the cipher processes it through ℓ parallel branches, *e.g.,* AIM2-V is depicted in Figure 1. The i-th branch $(1 \le i \le \ell)$ consists of the following operations:

- **Constant Addition (AC_1):** A fixed n-bit constant c_i is XORed with the key k.
- **Inverse S-box ($S^{-1}[e_i]$):** The non-linear layer is defined by exponentiation over $\mathbb{F}_{2^n}$. For $x \in \mathbb{F}_{2^n}$, $S^{-1}[e_i](x) = x^{(2^{e_i}-1)^{-1}}$, where e_i is a prime such that $2^{e_i} - 1$ is a Mersenne prime and $\gcd(e_i, n) = 1$.
- **Linear Layer (L):** The n-bit internal state is multiplied by a random invertible binary matrix $M_i \in \mathbb{F}_2^{n \times n}$, which is generated via an extendable-output function (XOF) seeded with a public initialization vector iv, and it can be equivalently expressed as a linearized polynomial L_i over $\mathbb{F}_{2^n}$.
- **Constant Addition (AC_2):** The internal state is XORed with another random n-bit constant c_i', also generated from XOF$[iv]$.

The outputs of these ℓ branches are then summed and processed through an additional S-box $S[e_{\ell+1}](x) = x^{2^{e_{\ell+1}}-1}$, where $\gcd(e_{\ell+1}, n) = 1$. Finally, a feed-forward operation is applied, rendering the overall function non-invertible. The designer instantiates three AIM2 variants for 128-/192-/256-bit security, the parameters of which are given in Table 2.

Table 2: Parameters sets of AIM2

Cipher	n	λ	ℓ	e_1	e_2	e_3	e_4	Irreducible Polynomial
AIM2-I	128	128	2	49	91	3	-	$X^{128} + X^7 + X^2 + X + 1$
AIM2-III	192	192	2	17	47	5	-	$X^{192} + X^7 + X^2 + X + 1$
AIM2-V	256	256	3	11	141	7	3	$X^{256} + X^{10} + X^5 + X^2 + 1$

3 More Accurate Security Evaluations of AIM2-III/V against Algebraic Attacks

The designers of AIM2 [22] showed that recovering the key of AIM2 from one (iv, ct) pair can be reduced to solving a large system of equations over $\mathbb{F}_2$. However, the system size is so large that useful algebraic structure is hidden. In this section, we instead work directly over $\mathbb{F}_{2^n}$, where we can use linearized polynomial decomposition [28] and carefully chosen eliminations to build smaller, lower-degree systems. In Section 3.1, we introduce a *resultant-minimized* model for AIM2-III, which reduces the number of resultants by combining S-box relations with low-degree linearized factors. This idea also applies to AIM2-V, however, in that case the very large exponents makes direct decomposition ineffective. To handle AIM2-V, we therefore propose a *subfield reduction* method in Section 3.2, making use of the fact that e_2 is close to 128, so that exponents can be reduced modulo a subfield. Together, these improvements give more accurate security evaluations of algebraic attacks, which also provide guidance for parameter selection and for strengthening designs. This helps us later in Section 4 to identify weak parameter choices of AIM2, and to propose a simple patched version AIM2-patch that mitigates our attacks with negligible efficiency loss.

3.1 Resultant-minimized Modeling for AIM2-III

Linearized Polynomial Decomposition. Given $l = 2$ parallel branches of AIM2-III (see Figure 2), the attack model is based on decomposing linearized polynomials $L_1(x) = \sum_{j=0}^{n-1} \gamma_{1,j} x^{2^j}$ and $L_2(x) = \sum_{j=0}^{n-1} \gamma_{2,j} x^{2^j}$, where $\gamma_{1,j}, \gamma_{2,j} \in \mathbb{F}_{2^n}$ are known coefficients. As shown in Proposition 1, the decomposition of $L_i(x)$ ($i \in \{1, 2\}$) reduces to finding a low-degree linearized polynomial $L_{b_3}(x)$ such that $L_{b_i}(x) = L_{b_3} \circ L_i(x)$ has degree at most 2^{t_i} for $i \in \{1, 2\}$, *i.e.*, $\deg(L_{b_i}) \leq 2^{t_i}$.

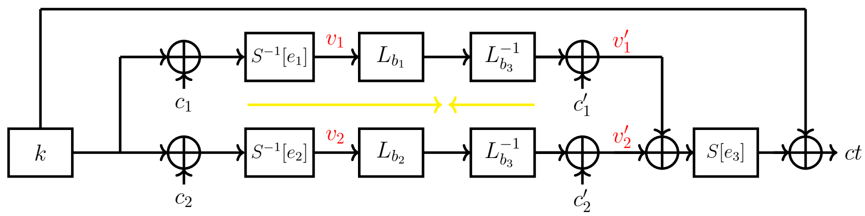

Fig. 2: Algebraic model of AIM2-III (intermediate variables are colored in red).

To minimize the number of variables, L_{b_1} and L_{b_2} are required to share the same polynomial $L_{b_3}(x) = \sum_{i=0}^{t_3} a_i x^{2^i}$, where the unknown coefficients $a_i \in \mathbb{F}_{2^n}$ are to be determined. All coefficients of the polynomial $L_{b_1} = \sum_{i=0}^{t_3} a_i (\sum_{j=0}^{n-1} \gamma_{1,j} x^{2^j})^{2^i}$ corresponding to terms of degree greater than t_1 are zero. Similarly, all terms in L_{b_2} with a degree exceeding t_2 have zero coefficients. Therefore, following [28], a system of $(2n - t_1 - t_2 - 2)$ equations in $(t_3 + 1)$ unknowns can be obtained:

$$\begin{cases} \sum\limits_{\substack{0 \le i \le t_3,\, 0 \le j \le n-1 \\ i+j \equiv \delta_1 \bmod n}} a_i \gamma_{1,j}^{2^i} = 0, \quad t_1 + 1 \le \delta_1 \le n-1, \\ \sum\limits_{\substack{0 \le i \le t_3,\, 0 \le j \le n-1 \\ i+j \equiv \delta_2 \bmod n}} a_i \gamma_{2,j}^{2^i} = 0, \quad t_2 + 1 \le \delta_2 \le n-1. \end{cases} \tag{5}$$

To ensure a nontrivial solution, one can let the number of variables exceed the number of equations, *i.e.*, $t_1 + t_2 + t_3 > 2n - 3$.

Resultant-minimized Modeling. We now present an attack model that is independent of the secret key k and minimizes the number of resultant computations. As illustrated in Figure 2, intermediate variables are introduced immediately after the S-box and the affine layer, which yields the following system

$$\begin{cases} v_1^{2^{e_1}-1} + v_2^{2^{e_2}-1} + c_1 + c_2 = 0, \\ (v_1' + v_2')^{2^{e_3}-1} + v_2^{2^{e_2}-1} + ct + c_2 = 0, \\ L_{b_1}(v_1) + L_{b_3}(v_1') + L_{b_3}(c_1') = 0, \\ L_{b_2}(v_2) + L_{b_3}(v_2') + L_{b_3}(c_2') = 0. \end{cases} \tag{6}$$

Then, setting $v_3 = v_1' + v_2'$ and $c = c_1' + c_2'$ and adding the last two equations gives the compact system as below

$$\begin{cases} g_1(v_1, v_2) := v_1^{2^{e_1}-1} + v_2^{2^{e_2}-1} + c_1 + c_2 = 0, \\ g_2(v_2, v_3) := v_3^{2^{e_3}-1} + v_2^{2^{e_2}-1} + ct + c_2 = 0, \\ f(v_1, v_2, v_3) := L_{b_1}(v_1) + L_{b_2}(v_2) + L_{b_3}(v_3) + L_{b_3}(c) = 0. \end{cases} \tag{7}$$

With this modeling, only two resultant eliminations (v_1 and v_3) are needed to obtain a univariate polynomial in v_2. Hence, this model has the minimum number of resultant operations for this MITM-style decomposition, thus reducing elimination cost compared to naïve formulations.

Optimized Non-k-based Substitution Strategy. During resultant elimination, the degrees of remaining variables grow rapidly. The fast-substitution technique in [28] controls this growth when a substitution of the form $x^{2^e-1} = k + c$ is available as follows.

Proposition 2 ([28]). *Let x^{2^ℓ} be a term appearing in a linearized polynomial $L(x)$, and suppose $x^{2^e-1} = k + c$, with c known. Write $\ell = se + r$ with unique integers $s \geq 0$ and $0 \leq r < e$. Then $x^{2^\ell} = x^{2^r} \cdot \prod_{j=1}^{s}\big(k^{2^{\ell-je}} + c^{2^{\ell-je}}\big)$. Let $L'(x,k)$ denote $L(x)$ after applying this substitution, then $\deg_x(L'(x,k)) \leq 2^{e-1}, \deg_k(L'(x,k)) < 2^{\ell-e+1}$.*

In this paper, we adopt a non-k-based substitution strategy and apply the substitution with v_2 as the base in Equation (7) (which is optimal in our modeling). This can reduce the size of the determinant in the resultant computation, where the determinant size has a significant impact on the complexity. In particular, using

$$v_1^{2^{e_1}} = v_1 \cdot \big(v_2^{2^{e_2}-1} + c_1 + c_2\big)$$

in $L_{b_1}(v_1)$ yields the substituted polynomial $L'_{b_1}(v_1, v_2)$. Writing $t_1 = s_1 e_1 + r_1$ with $0 \leq r_1 < e_1$ gives

$$v_1^{2^{t_1}} = v_1^{2^{r_1}} \cdot \prod_{j=1}^{s_1}\Big((v_2^{2^{e_2}-1})^{2^{\,t_1-je_1}} + (c_1+c_2)^{2^{\,t_1-je_1}}\Big). \tag{8}$$

Consequently, $\deg_{v_1}(L'_{b_1}) \leq 2^{e_1} - 1$. Similarly, applying the substitution $v_3^{2^{e_3}} = v_3 \cdot (v_2^{2^{e_2}-1} + ct + c_2)$ to $L_{b_3}(v_3)$ and obtaining $L'_{b_3}(v_3, v_2)$ with $\deg_{v_3}(L'_{b_3}) \leq 2^{e_3} - 1$. Moreover, it has the degree bounds after substitution

$$\deg_{v_2}\big(L'_{b_1}\big) < 2^{\,t_1-e_1+e_2+1}, \qquad \deg_{v_2}\big(L'_{b_3}\big) < 2^{\,t_3-e_3+e_2+1}.$$

Hence, the substituted system becomes

$$\begin{cases} g_1 := v_1^{2^{e_1}-1} + v_2^{2^{e_2}-1} + c_1 + c_2 = 0, \\ g_2 := v_3^{2^{e_3}-1} + v_2^{2^{e_2}-1} + ct + c_2 = 0, \\ f' := L'_{b_1}(v_1, v_2) + L_{b_2}(v_2) + L'_{b_3}(v_3, v_2) + L_{b_3}(c) = 0, \end{cases} \tag{9}$$

with

$$\deg_{v_2}(f') \leq \max\big(2^{t_1-e_1+e_2+1},\ 2^{t_2},\ 2^{t_3-e_3+e_2+1}\big). \tag{10}$$

These bounds control degree growth during successive resultant eliminations and are central to our complexity evaluations.

Elimination and Derivation of the Univariate Polynomial. We now perform resultant elimination on Equation (9) as below.

Step 1. Firstly, we compute $G_1(v_1, v_2) = \mathrm{Res}_{v_3}(g_2, f')$ to eliminate v_3. By Definition 2, the Sylvester matrix has size at most $(3\cdot 2^{e_3-1}-1)\times(3\cdot 2^{e_3-1}-1)$. Entries in the first 2^{e_3-1} rows have $\deg_{v_1} = 0$ and $\deg_{v_2} = 2^{e_2}-1$, entries in the remaining $2^{e_3}-1$ rows have $\deg_{v_1} = 2^{e_1-1}$ and $\deg_{v_2} = \deg_{v_2}(f')$. Hence, it has

$$\deg_{v_1}(G_1) \le (2^{e_3}-1)\cdot 2^{e_1-1} < 2^{e_1+e_3-1},$$

$$\deg_{v_2}(G_1) \le 2^{e_3-1}(2^{e_2}-1) + (2^{e_3}-1)\deg_{v_2}(f').$$

Apply the substitution $v_1^{2^{e_1}} = v_1\left(v_2^{2^{e_2}-1} + c_1 + c_2\right)$ to $G_1(v_1, v_2)$ and obtain $G_1'(v_1, v_2)$ with $\deg_{v_1}(G_1') \le 2^{e_1}-2$. Then using $v_1^{2^{e_1+e_3-1}} = v_1^{2^{e_3-1}}\left(v_2^{2^{e_2}-1} + c_1 + c_2\right)^{2^{e_3-1}}$, substituting the highest-degree cross term of G_1 yields

$$\deg_{v_2}(G_1') \le 2^{e_3-1}(2^{e_2}-1) + \deg_{v_2}(G_1).$$

Step 2. Secondly, we compute $G_2(v_2) = \mathrm{Res}_{v_1}(g_1, G_1')$ to eliminate v_1. The corresponding Sylvester matrix has size at most $(2^{e_1+1}-3)\times(2^{e_1+1}-3)$. Entries in the first $2^{e_1}-2$ rows have $\deg_{v_2} = 2^{e_2}-1$, and entries in the remaining $2^{e_1}-1$ rows have $\deg_{v_2} = \deg_{v_2}(G_1')$. Thus, it has

$$\deg_{v_2}(G_2) \le (2^{e_1}-2)(2^{e_1}-1) + (2^{e_1}-1)\deg_{v_2}(G_1').$$

Step 3. Finally, we solve the resulting univariate polynomial $G_2(v_2) = 0$.

Complexity Analysis. The construction of L_{b_3} for decomposition can be precomputed in time $\mathcal{O}(n^\omega)$, which is negligible here. The substitution in Equation (8) requires computing $\prod_{j=1}^{s_1}\left((v_2^{2^{e_2}-1})^{2^{t_1-je_1}} + (c_1+c_2)^{2^{t_1-je_1}}\right)$ with $s_1 \le \lceil t_1/e_1\rceil \le \lceil n/e_1\rceil$. This costs at most $\mathcal{O}(2^{s_1})$ field operations, and since $s_1 \ll t_1$, the substitution cost is negligible. Concretely, the substitution step is bounded by $\mathcal{O}(t_1 2^{s_1} + t_3 2^{s_3})$ field operations, which is small relative to the elimination costs. For resultant elimination, we follow the framework in [6,24] by evaluating a $D_H \times D_H$ determinant of entries of degree up to d costs $\mathcal{O}\left(D_H^\omega\,\mathcal{M}(d)\right)$, where $\mathcal{M}(d)$ is the multivariate multiplication cost in Equation (3).

- In Step 1, for $G_1 = \mathrm{Res}_{v_3}(g_2, f')$, the Sylvester matrix has size at most $(3\cdot 2^{e_3-1}-1)\times(3\cdot 2^{e_3-1}-1)$, which can be reduced to $(2^{e_3}-1)\times(2^{e_3}-1)$ by applying elementary row operations (following the approach in [28]). Concretely, we apply Gaussian elimination to the last $2^{e_3}-1$ rows, using unit pivots from the first 2^{e_3-1} rows, and reduce the Sylvester matrix to size $(2^{e_3}-1)\times(2^{e_3}-1)$. In the reduced matrix, each entry has $\deg_{v_1} \le 2^{e_1-1}$ and $\deg_{v_2} \le 2^{e_2}-1+\deg_{v_2}(f')$. Hence, the cost of this step is

$$T_1 = \mathcal{O}((2^{e_3}-1)^\omega\,\mathcal{M}(d_1)), \qquad d_1 := 2^{e_1-1}\left(2^{e_2}-1+\deg_{v_2}(f')\right).$$

– In Step 2, for $G_2 = \mathrm{Res}_{v_1}(g_1, G_1')$, the Sylvester matrix can similarly be reduced to size $(2^{e_1}-1)\times(2^{e_1}-1)$, with entry degree bounded by $2^{e_2}-1+\deg_{v_2}(G_1')$. Thus

$$T_2 = \mathcal{O}\big((2^{e_1}-1)^{\omega}\,\mathcal{M}(d_2)\big), \qquad d_2 := 2^{e_2}-1+\deg_{v_2}(G_1').$$

– In Step 3, solving the univariate polynomial $G_2(v_2)$ costs $T_3 = \mathcal{U}\big(\deg_{v_2}(G_2)\big)$, where $\mathcal{U}(\cdot)$ is the root-finding cost in Equation (4).

Finally, the overall attack time complexity is $T = T_1 + T_2 + T_3$.

Attack Parameters. The costs of the resultant computations and root-finding for $G_2(v_2)$ heavily depend on $\deg_{v_2}(f')$, and according to Equation (10), one should choose t_1, t_2, t_3 to minimize $\max\big(t_1-e_1+e_2+1,\ t_2,\ t_3-e_3+e_2+1\big)$, which is subject to the feasibility constraint $t_1+t_2+t_3 > 2n-3$. In practice, we search for low-degree decompositions of the linearized factors and pick parameters that balance the three terms above, and obtain $t_1 = 121, t_2 = 152, t_3 = 109$ for $L_1 = L_{b_1} \circ L_{b_3}^{-1}$ and $L_2 = L_{b_2} \circ L_{b_3}^{-1}$. Under these attack parameters, and with $\omega = 2$ ($\omega = 2.373$), it has

$$\deg_{v_2}(f') = 2^{152}, \qquad d_1 < 2^{169}, \qquad d_2 < 2^{158}, \qquad \deg_{v_2}(G_2) < 2^{175}.$$

According to the complexity analysis above, it leads to the overall attack time complexity $T = T_1+T_2+T_3 \approx 2^{201}$ ($2^{207.29}$) field operations, which is equivalent to $2^{188.76}$ ($2^{195.05}$) primitive calls.

3.2 Subfield-based Reduction with Application to AIM2-V

The above attack for AIM2-III naturally can extend to AIM2-V, however, a large exponent e_2 makes the degrees of basis variables (after substitution) too high, thus rendering linearized-polynomial decomposition infeasible. To address this, we consider building the equation system over a subfield, *e.g.*, $\mathbb{F}_{2^{128}}$ for AIM2-V defined over $\mathbb{F}_{2^{256}}$, which could effectively reduce the degrees of equations.

Resultant-minimized Modeling. Similar to the decomposition in Section 3.1, as shown in Figure 3, we let $L_{b_i} = L_{b_4} \circ L_i$ for $1 \le i \le 3$, and introduce intermediate variables v_i after $S^{-1}[e_i]$ and v_i' after the affine layer. Then, the MITM-style system for AIM2-V is

$$\begin{cases} v_2^{2^{e_2}-1} + v_1^{2^{e_1}-1} + c_2 + c_1 = 0, \\ v_3^{2^{e_3}-1} + v_1^{2^{e_1}-1} + c_3 + c_1 = 0, \\ (v_1' + v_2' + v_3')^{2^{e_4}-1} + v_1^{2^{e_1}-1} + ct + c_1 = 0, \\ L_{b_1}(v_1) + L_{b_4}(v_1') + L_{b_4}(c_1') = 0, \\ L_{b_2}(v_2) + L_{b_4}(v_2') + L_{b_4}(c_2') = 0, \\ L_{b_3}(v_3) + L_{b_4}(v_3') + L_{b_4}(c_3') = 0. \end{cases} \tag{11}$$

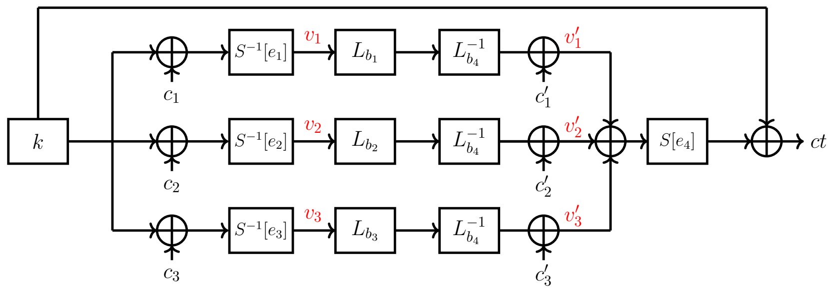

Fig. 3: Algebraic model of AIM2-V (intermediate variables are colored in red).

Setting $v_4 = v_1' + v_2' + v_3'$ and $c_4 = c_1' + c_2' + c_3'$, and adding the last three affine equations yields the compact system

$$\begin{cases} v_2^{2^{e_2}} + v_2\big(v_1^{2^{e_1}-1} + c_2 + c_1\big) = 0, \\ v_3^{2^{e_3}} + v_3\big(v_1^{2^{e_1}-1} + c_3 + c_1\big) = 0, \\ v_4^{2^{e_4}} + v_4\big(v_1^{2^{e_1}-1} + ct + c_1\big) = 0, \\ L_{b_1}(v_1) + L_{b_2}(v_2) + L_{b_3}(v_3) + L_{b_4}(v_4) + L_{b_4}(c_4) = 0. \end{cases} \tag{12}$$

Note that Equation (12) is the starting point for the subfield reduction, by expressing elements of $\mathbb{F}_{2^{256}}$ in the basis of the quadratic extension $\mathbb{F}_{2^{256}} \cong \mathbb{F}_{2^{128}}[\alpha]$ (with α is the generator of the extension), each equation can be decomposed into two equations over $\mathbb{F}_{2^{128}}$, This helps to reduce large exponents modulo $2^{128}-1$ and produce much lower-degree systems amenable to resultant elimination or other algebraic techniques, as detailed below.

Linearized Polynomial Decomposition. We now derive a linear decomposition of Equation (12) tailored for the subfield reduction later. Let

$$L_{b_4}(x) = \sum_{i=0}^{255} a_i x^{2^i}, L_1(x) = \sum_{j=0}^{255} \gamma_{1,j} x^{2^j}, L_2(x) = \sum_{j=0}^{255} \gamma_{2,j} x^{2^j}, L_3(x) = \sum_{j=0}^{255} \gamma_{3,j} x^{2^j},$$

where $\gamma_{\ell,j} \in \mathbb{F}_{2^{256}}$ are known and $a_i \in \mathbb{F}_{2^{256}}$ are unknown coefficients. Then, $L_{b_1}(x)$, $L_{b_2}(x)$ and $L_{b_3}(x)$ are as below.

$$\begin{cases} L_{b_1}(x) = L_{b_4}(L_1(x)) = \sum_{i=0}^{255} a_i (\sum_{j=0}^{255} \gamma_{1,j} x^{2^j})^{2^i} \\ L_{b_2}(x) = L_{b_4}(L_2(x)) = \sum_{i=0}^{255} a_i (\sum_{j=0}^{255} \gamma_{2,j} x^{2^j})^{2^i} \\ L_{b_3}(x) = L_{b_4}(L_3(x)) = \sum_{i=0}^{255} a_i (\sum_{j=0}^{255} \gamma_{3,j} x^{2^j})^{2^i} \end{cases}$$

To exploit the subfield $\mathbb{F}_{2^{128}} \subset \mathbb{F}_{2^{256}}$, each linearized polynomial is divided into low and high 128-bit parts, as depicted in Figure 4. Concretely, we carefully

$$\underbrace{\boxed{a_0x + a_1x^2 + \cdots + a_{t_4}x^{2^{t_4}} + 0x^{2^{t_4+1}} + \cdots + 0x^{2^{127}}} + \boxed{a_{128}x^{2^{128}} + \cdots + a_{128+t_4}x^{2^{128+t_4}} + 0x^{2^{129+t_4}} + \cdots + 0x^{2^{255}}}}_{\deg(L_{b_4}) = 2^{128+t_4} \quad \text{over} \quad \mathbb{F}_{2^{256}}}$$

$$\downarrow$$

$$\underbrace{\begin{array}{c} \boxed{a_0x + a_1x^2 + \cdots + a_{t_4}x^{2^{t_4}}} \\ + \\ \boxed{a_{128}x + a_{129}x^2 + \cdots + a_{128+t_4}x^{2^{t_4}}} \end{array} \quad \Longrightarrow \quad (a_0 + a_{128})x + (a_1 + a_{129})x^2 + \cdots + (a_{t_4} + a_{128+t_4})x^{2^{t_4}}}_{\deg(L_{b_4}) = 2^{t_4} \quad \text{over} \quad \mathbb{F}_{2^{128}}}$$

Fig. 4: The representations of $L_{b_4}(x)$ over $\mathbb{F}_{2^{256}}$ and $\mathbb{F}_{2^{128}}$.

choose integers $0 \le t_1, t_2, t_3, t_4 \le 128$ and derive the following

$$\begin{cases} L_{b_4}(x) &= \sum_{i=0}^{t_4} a_i x^{2^i} \ + \ \sum_{i=128}^{128+t_4} a_i x^{2^i}, \\ L_{b_\eta}(x) &= \sum_{\delta=0}^{t_\eta} \left(\sum_{\substack{0 \le i,j \le 255 \\ i+j \equiv \delta \pmod{256}}} a_i \gamma_{\eta,j}^{2^i} \right) x^{2^\delta} + \sum_{\delta=128}^{128+t_\eta} \left(\sum_{\substack{0 \le i,j \le 255 \\ i+j \equiv \delta \pmod{256}}} a_i \gamma_{\eta,j}^{2^i} \right) x^{2^\delta}, \end{cases}$$

for $\eta = 1, 2, 3$. We require the highest nonzero coefficients in each 128-bit half, *i.e.*, $a_{t_4}, a_{t_4+128} \neq 0$, and assume $\sum_{i+j \equiv t_\eta} a_i \gamma_{\eta,j}^{2^i} \neq 0$ and $\sum_{i+j \equiv t_\eta + 128} a_i \gamma_{\eta,j}^{2^i} \neq 0$. Then, we collect the forced-zero coefficients a_i to obtain $2 \cdot \big(3(128-1) - t_1 - t_2 - t_3\big)$ linear equations in the $2(t_4+1)$ unknowns $a_0, \ldots, a_{t_4}, a_{128}, \ldots, a_{128+t_4}$ as below

$$\begin{cases} a_i = 0 & \text{for } t_4 + 1 \le i \le 127 \text{ and } t_4 + 129 \le i \le 255, \\ \displaystyle\sum_{\substack{0 \le i,j \le 255 \\ i+j \equiv \delta_1 \pmod{256}}} a_i \gamma_{1,j}^{2^i} = 0 & \text{for } t_1 + 1 \le \delta_1 \le 127 \text{ and } t_1 + 129 \le \delta_1 \le 255, \\ \displaystyle\sum_{\substack{0 \le i,j \le 255 \\ i+j \equiv \delta_2 \pmod{256}}} a_i \gamma_{2,j}^{2^i} = 0 & \text{for } t_2 + 1 \le \delta_2 \le 127 \text{ and } t_2 + 129 \le \delta_2 \le 255, \\ \displaystyle\sum_{\substack{0 \le i,j \le 255 \\ i+j \equiv \delta_3 \pmod{256}}} a_i \gamma_{3,j}^{2^i} = 0 & \text{for } t_3 + 1 \le \delta_3 \le 127 \text{ and } t_3 + 129 \le \delta_3 \le 255. \end{cases}$$

To admit a nontrivial solution, it hence requires $t_1 + t_2 + t_3 + t_4 > 3 \cdot 128 - 4$.

With the above decomposition, we get back to Equation (12) with the degrees of each equation over $\mathbb{F}_{2^{256}}$ as below

- deg of the first equation: 2^{141} ($e_2 = 141$ for AIM2-V),
- deg of the second and third equations: 2^{11} ($e_1 = 11$ for AIM2-V),
- deg of the fourth equation: $\max(2^{128+t_1}, 2^{128+t_2}, 2^{128+t_3}, 2^{128+t_4})$.

Next, all field elements will be expressed in the basis of the quadratic extension $\mathbb{F}_{2^{256}} \cong \mathbb{F}_{2^{128}}(\alpha)$, so that each equation splits into two equations over $\mathbb{F}_{2^{128}}$, thus reducing large exponents modulo $2^{128} - 1$ and producing a much lower-degree system suitable for elimination.

Subfield Reduction. Define the field extension $\mathbb{F}_{2^{256}} \cong \mathbb{F}_{2^{128}}[X]/(X^2+X+\beta)$, where $\beta \in \mathbb{F}_{2^{128}}$ and $X^2+X+\beta$ is irreducible over $\mathbb{F}_{2^{128}}$. Let α be a root of $X^2+X+\beta$, *i.e.*, $\alpha^2 = \alpha+\beta$. Every element and equation over $\mathbb{F}_{2^{256}}$ admits a unique decomposition into subfield coordinates as detailed below.

Proposition 3 ([25]). *Let $(1,\alpha)$ be a basis of $\mathbb{F}_{2^{256}}$ over $\mathbb{F}_{2^{128}}$. Then*

1. *Every $x \in \mathbb{F}_{2^{256}}$ can be written uniquely as $x = a+\alpha b$, where $a, b \in \mathbb{F}_{2^{128}}$.*
2. *If a polynomial $F(x) \in \mathbb{F}_{2^{256}}[x]$ decomposes as $F(x) = F_1(x_1,x_2)+\alpha F_2(x_1,x_2)$ with $F_1, F_2 \in \mathbb{F}_{2^{128}}[x_1,x_2]$ and $x = x_1+\alpha x_2$, then*

$$F(x) = 0 \iff F_1(x_1,x_2) = 0 \text{ and } F_2(x_1,x_2) = 0.$$

Now, for Equation (12), according to Proposition 3, we let $v_i = v_{i,1} + \alpha v_{i,2}, c_i = c_{i,1} + \alpha c_{i,2}, ct = ct_1 + \alpha ct_2$, where $v_{i,j}, c_{i,j}, ct_j \in \mathbb{F}_{2^{128}}$ for $1 \le i \le 4$, $1 \le j \le 2$. Then, it has the following equations over $\mathbb{F}_{2^{128}}$

$$\begin{cases} (v_{2,1}+\alpha v_{2,2})^{2^{e_2}} = (v_{2,1}+\alpha v_{2,2}) \cdot \left((v_{1,1}+\alpha v_{1,2})^{2^{e_1}-1} + c_{1,1} + c_{2,1} + \alpha(c_{1,2}+c_{2,2})\right), \\ (v_{3,1}+\alpha v_{3,2})^{2^{e_3}} = (v_{3,1}+\alpha v_{3,2}) \cdot \left((v_{1,1}+\alpha v_{1,2})^{2^{e_1}-1} + c_{1,1} + c_{3,1} + \alpha(c_{1,2}+c_{3,2})\right), \\ (v_{4,1}+\alpha v_{4,2})^{2^{e_4}} = (v_{4,1}+\alpha v_{4,2}) \cdot \left((v_{1,1}+\alpha v_{1,2})^{2^{e_1}-1} + c_{1,1} + ct_1 + \alpha(c_{1,2}+ct_2)\right), \\ \sum_{i=1}^{4} L_{b_i}(v_{i,1}+\alpha v_{i,2}) + L_{b_4}(c_{4,1}+\alpha c_{4,2}) = 0. \end{cases} \tag{13}$$

Decompose the first three equations over $\mathbb{F}_{2^{128}}$. By expressing all variables and constants in the basis $(1,\alpha)$, each of the first three equations in Equation (13) splits into two component equations over $\mathbb{F}_{2^{128}}$. In total, we obtain six equations:

$$\begin{cases} g_{i,1} := v_{i,1}^{2^{e_i}} + v_{i,1} f_{i,1}(v_{1,1},v_{1,2}) + v_{i,2} f_{i,2}(v_{1,1},v_{1,2}) = 0, \\ g_{i,2} := v_{i,2}^{2^{e_i}} + v_{i,1} f_{i,3}(v_{1,1},v_{1,2}) + v_{i,2} f_{i,4}(v_{1,1},v_{1,2}) = 0, \end{cases} \tag{14}$$

where for $2 \le i \le 4$ and $1 \le j \le 4$, and it has $\deg_{v_{1,1}}(f_{i,j}) = \deg_{v_{1,2}}(f_{i,j}) \le 2^{e_1}-1$. The details of decomposition are provided in Appendix A. For example, for the first equation of Equation (12), it has the degree 2^{141} over $\mathbb{F}_{2^{256}}$, while after rewriting over $\mathbb{F}_{2^{128}}$, it turns into two equations $g_{2,1}$ and $g_{2,2}$ of degree 2^{13} only, due to the subfield reduction $x^{2^{e_2}} = x^{2^{e_2-128}}$. Thus, *the effective exponent e_2 is reduced from 141 to 13, significantly lowering the degree of the system.*

Decompose the last equation over $\mathbb{F}_{2^{128}}$. As illustrated in Figure 4, the linearized polynomial L_{b_4} has a structural symmetry, *i.e.*, the highest nonzero terms in both low and high 128-bit segments occupy the same relative positions. Owing to this symmetry, the degree of L_{b_4} over the subfield is exactly t_4. Then, let

$a_i = a_{i,1} + \alpha a_{i,2}$ and $\alpha^{2^i} = \alpha + C_i$[8], $L_{b_4}(v_4)$ can be expressed over $\mathbb{F}_{2^{128}}$ as

$$
\begin{aligned}
L_{b_4}(v_4) &= \sum_{i=0}^{t_4}(a_{i,1}+\alpha a_{i,2})(v_{4,1}+\alpha v_{4,2})^{2^i} + \sum_{i=128}^{128+t_4}(a_{i,1}+\alpha a_{i,2})(v_{4,1}+\alpha v_{4,2})^{2^i} \\
&= \sum_{i=0}^{t_4}\big(a_{i,1}+a_{i+128,1}+\alpha(a_{i,2}+a_{i+128,2})\big)(v_{4,1}+\alpha v_{4,2})^{2^i} \\
&= \sum_{i=0}^{t_4}\Big[(a_{i,1}+a_{i+128,1})v_{4,1}^{2^i} + \big(C_i(a_{i,1}+a_{i+128,1})+C_1(a_{i,2}+a_{i+128,2})\big)v_{4,2}^{2^i}\Big] \\
&\quad + \alpha\sum_{i=0}^{t_4}\Big[(a_{i,2}+a_{i+128,2})v_{4,1}^{2^i} + \big(a_{i,1}+a_{i+128,1}+C_i(a_{i,2}+a_{i+128,2})\big)v_{4,2}^{2^i}\Big] \\
&= L_{b_{4,1}}(v_{4,1}) + L_{b_{4,2}}(v_{4,2}) + \alpha\big(L_{b_{4,3}}(v_{4,1}) + L_{b_{4,4}}(v_{4,2})\big),
\end{aligned}
$$

where $L_{b_{4,1}}$, $L_{b_{4,2}}$, $L_{b_{4,3}}$, and $L_{b_{4,4}}$ are linearized polynomials of degree t_4 over $\mathbb{F}_{2^{128}}$. Applying the similar decomposition to $L_{b_1}(v_1)$, $L_{b_2}(v_2)$, $L_{b_3}(v_3)$, and $L_{b_4}(c_4)$ yields two subfield equations

$$
\begin{cases}
F_1 := \displaystyle\sum_{i=1}^{4}\sum_{j=1}^{2} L_{b_{i,j}}(v_{i,j}) + L_{b_{4,1}}(c_{4,1}) + L_{b_{4,2}}(c_{4,2}) = 0, \\
F_2 := \displaystyle\sum_{i=1}^{4}\sum_{j=1}^{2} L_{b_{i,j+2}}(v_{i,j}) + L_{b_{4,3}}(c_{4,1}) + L_{b_{4,4}}(c_{4,2}) = 0.
\end{cases} \tag{15}
$$

Then, for $1 \le i \le 4$ and $1 \le j \le 2$, it has $\deg_{v_{i,j}}(F_1) = \deg_{v_{i,j}}(F_2) \le 2^{t_i}$. In total, the AIM2-V description over $\mathbb{F}_{2^{128}}$ consists of eight equations in eight variables $v_{i,j}$ ($1 \le i \le 4$, $1 \le j \le 2$) as below

- two equations of degree 2^{13} from Equation (14) with $i = 2$,
- four equations of degree 2^{11} from Equation (14) with $i = 3, 4$,
- two equations of degree $\max(2^{t_1}, 2^{t_2}, 2^{t_3}, 2^{t_4})$ from Equation (15).

Substitution over $\mathbb{F}_{2^{128}}$. For $2 \le i \le 4$, *i.e.*, Equation (14), we apply the substitutions

$$v_{i,1}^{2^{e_i}} = v_{i,1}\, f_{i,1}(v_{1,1}, v_{1,2}) + v_{i,2}\, f_{i,2}(v_{1,1}, v_{1,2})$$

to $L_{b_{i,1}}(v_{i,1})$ and $L_{b_{i,3}}(v_{i,1})$, and denote the resulting polynomials by $L'_{b_{i,1}}$ and $L'_{b_{i,3}}$, respectively. For the term $v_{i,1}^{2^{t_i}}$, we apply the substitution

$$
\begin{aligned}
v_{i,1}^{2^{t_i}} &= \big(v_{i,1}^{2^{e_i}}\big)^{2^{t_i-e_i}} \\
&= v_{i,1}^{2^{t_i-e_i}}\, f_{i,1}^{2^{t_i-e_i}}(v_{1,1}, v_{1,2}) + v_{i,2}^{2^{t_i-e_i}}\, f_{i,2}^{2^{t_i-e_i}}(v_{1,1}, v_{1,2}) \\
&= v_{i,1}^{2^{r_i}} \cdot \prod_{j=1}^{s_i} f_{i,1}^{2^{t_i-je_i}}(v_{1,1}, v_{1,2}) + v_{i,2}^{2^{r_i}} \cdot \prod_{j=1}^{s_i} f_{i,2}^{2^{t_i-je_i}}(v_{1,1}, v_{1,2}),
\end{aligned} \tag{16}
$$

[8] $\alpha^{2^i} = (\alpha+\beta)^{2^{i-1}} = (\alpha+\beta+\beta^2)^{2^{i-2}} = \cdots = \alpha + C_i$, where C_i is a polynomial of β. Specifically, $C_1 = \beta$.

where $t_i = s_i e_i + r_i$ with $0 \le r_i < e_i$. Consequently,

$$\deg_{v_{i,1}}\left(L'_{b_{i,1}}\right) = \deg_{v_{i,1}}\left(L'_{b_{i,3}}\right) \le 2^{e_i-1}, \deg_{v_{1,1}}\left(L'_{b_{i,1}}\right) = \deg_{v_{1,1}}\left(L'_{b_{i,3}}\right) \le 2^{\,t_i-e_i+e_1+1}.$$

The substitutions for $v_{i,2}$ are analogous, and we denote

$$D := \max\bigl(2^{t_1},\, 2^{t_2-e_2+e_1+1},\, 2^{t_3-e_3+e_1+1},\, 2^{t_4-e_4+e_1+1}\bigr). \tag{17}$$

After substitution, we then obtain

$$\begin{cases} g_{i,1} := v_{i,1}^{2^{e_i}} + v_{i,1} f_{i,1}(v_{1,1}, v_{1,2}) + v_{i,2} f_{i,2}(v_{1,1}, v_{1,2}) = 0, \text{ for } 2 \le i \le 4, \\ g_{i,2} := v_{i,2}^{2^{e_i}} + v_{i,1} f_{i,3}(v_{1,1}, v_{1,2}) + v_{i,2} f_{i,4}(v_{1,1}, v_{1,2}) = 0, \text{ for } 2 \le i \le 4, \\ F'_1 := \sum_{j=1}^{2} L_{b_{1,j}}(v_{1,j}) + \sum_{i=2}^{4} \sum_{j=1}^{2} L'_{b_{i,j}} + L_{b_{4,1}}(c_{4,1}) + L_{b_{4,2}}(c_{4,2}) = 0, \\ F'_2 := \sum_{j=1}^{2} L_{b_{1,j+2}}(v_{1,j}) + \sum_{i=2}^{4} \sum_{j=1}^{2} L'_{b_{i,j+2}} + L_{b_{4,3}}(c_{4,1}) + L_{b_{4,4}}(c_{4,2}) = 0. \end{cases} \tag{18}$$

where $\deg_{v_{1,1}}(F'_1)$, $\deg_{v_{1,2}}(F'_1)$, $\deg_{v_{1,1}}(F'_2)$ and $\deg_{v_{1,2}}(F'_2)$ are bounded by D.

Resultant Elimination. We perform resultant eliminations on Equation (18) and fix the elimination order as $v_{2,1} \succ v_{2,2} \succ v_{3,1} \succ v_{3,2} \succ v_{4,1} \succ v_{4,2}$.

First elimination. We compute three resultants

$$Res(g_{2,1}, g_{2,2}, v_{2,1}),\ Res(g_{2,1}, F'_1, v_{2,1}),\ Res(g_{2,1}, F'_2, v_{2,1})$$

to eliminate $v_{2,1}$ and update

$$g' \leftarrow Res(g_{2,1}, g_{2,2}, v_{2,1}), \quad G_1 \leftarrow Res(g_{2,1}, F'_1, v_{2,1}), \quad G_2 \leftarrow Res(g_{2,1}, F'_2, v_{2,1}).$$

Then, apply the substitutions

$$\begin{cases} v_{i,1}^{2^{e_i}} = v_{i,1} f_{i,1}(v_{1,1}, v_{1,2}) + v_{i,2} f_{i,2}(v_{1,1}, v_{1,2}), \\ v_{i,2}^{2^{e_i}} = v_{i,1} f_{i,3}(v_{1,1}, v_{1,2}) + v_{i,2} f_{i,4}(v_{1,1}, v_{1,2}), \end{cases} \qquad i \in \{3, 4\},$$

to G_1 and G_2, and write the substituted polynomials as G'_1 and G'_2.

Second elimination. We compute two resultants

$$Res(g', G'_1, v_{2,2}),\ Res(g', G'_2, v_{2,2})$$

to eliminate $v_{2,2}$ and update

$$G_3 \leftarrow Res(g', G'_1, v_{2,2}), \qquad G_4 \leftarrow Res(g', G'_2, v_{2,2}).$$

Then, apply the same substitutions (for $i \in \{3, 4\}$) to G_3 and G_4 to obtain G'_3 and G'_4, which are updated to $F'_1 \leftarrow G'_3, F'_2 \leftarrow G'_4$.

Proceeding similarly and eliminating $v_{2,1}$, $v_{2,2}$, $v_{3,1}$, $v_{3,2}$, $v_{4,1}$ and $v_{4,2}$ yields the generators, from which we finally obtain two bivariate polynomials in $v_{1,1}$ and $v_{1,2}$ as given in Algorithm 1.

Algorithm 1 Obtain Bivariate Polynomials for $v_{1,1}$ and $v_{1,2}$

Input: $g_{i,j}$ $(2 \le i \le 4,\ 1 \le j \le 2)$, F_1', F_2' as in Equation (18).
Output: Two bivariate polynomials in $(v_{1,1}, v_{1,2})$.

for all $i \in \{2, 3, 4\}$ **do**
  $g' \leftarrow Res(g_{i,1}, g_{i,2}, v_{i,1})$;
  $G_1 \leftarrow Res(g_{i,1}, F_1', v_{i,1})$, $G_2 \leftarrow Res(g_{i,1}, F_2', v_{i,1})$;
  Apply the substitutions to G_1, G_2 and obtain G_1', G_2';
  $G_3 \leftarrow Res(g', G_1', v_{i,2})$, $G_4 \leftarrow Res(g', G_2', v_{i,2})$;
  Apply the substitutions to G_3, G_4 and obtain G_3', G_4';
  $F_1' \leftarrow G_3'$, $F_2' \leftarrow G_4'$;
end for
return F_1', F_2'.

Final step. Rather than a last resultant, we guess $v_{1,1}$ and solve for $v_{1,2}$ with a half-gcd routine, which is negligible compared with the eliminations above.

Complexity Analysis. The costs of constructing L_{b_4} and performing the substitutions are negligible compared to the eliminations. The complexity of each resultant step is governed by the Sylvester determinant size and the degrees of the surviving variables. In Appendix B, we track the degree growth of $v_{1,1}$ and $v_{1,2}$ through elimination and give bounds for removing $v_{2,1}$, $v_{2,2}$, $v_{3,1}$, $v_{3,2}$, $v_{4,1}$, and $v_{4,2}$, the complexities of which are denoted by $T_1, \ldots, T_6$. Finally, from the two bivariate equations F_1', F_2' returned by Algorithm 1, we guess $v_{1,1}$ and solve for $v_{1,2}$ via a half-gcd routine with the cost

$$T_7 \;=\; 2^{128} \cdot \deg_{v_{1,1}}(F_1') \, \log\bigl(\deg_{v_{1,1}}(F_1')^2\bigr).$$

The overall time complexity is therefore $T = \sum_{i=1}^{7} T_i$.

Parameter Selection. The dominant factor is D in Equation (17), hence we choose t_1, t_2, t_3, t_4 to minimize $\max(t_1, t_2-e_2+e_1+1, t_3-e_3+e_1+1, t_4-e_4+e_1+1)$ under the constraint $t_1+t_2+t_3+t_4 > 3 \cdot 128 - 4$. We randomly sampled three linearized permutation polynomials L_1, L_2, L_3 and obtain $t_1 = 99, t_2 = 99, t_3 = 93, t_4 = 90$ for $L_1 = L_{b_1} \circ L_{b_4}^{-1}, L_2 = L_{b_2} \circ L_{b_4}^{-1}, L_3 = L_{b_3} \circ L_{b_4}^{-1}$. We now proceed to evaluate the complexity for this parameter setting.

Using the substitution method proposed before, we have

$$\deg_{v_{2,j}}(F_1^{'}) < 2^{13}-1, \quad \deg_{v_{3,j}}(F_1^{'}) < 2^{7}-1, \quad \deg_{v_{4,j}}(F_1^{'}) < 2^{3}-1, \quad j \in \{1,2\}.$$

Their degrees in $F_2^{'}$ are likewise equal. First, we give the concrete complexity of the first two Resultant computation. According to the complexity analysis in Appendix B, under these attack parameters, and for $\omega = 2$ ($\omega = 2.373$), we have

$$D = 2^{99}, \qquad d_1 < 2^{231}, \qquad d_2 < 2^{35}, \qquad d_3 < 2^{244},$$

where d_i denotes the degree of the remaining variables in multivariate polynomial multiplication. In this setting, $T_1 \approx 2^{270.12}$ ($2^{275.23}$) field operations, which is equivalent to $2^{257.12}$ ($2^{262.23}$) primitive calls. Similarly, $T_2 \approx 2^{308.97}$ ($2^{318.81}$) field operations, equivalent to about $2^{295.97}$ ($2^{305.81}$) primitive calls. The computations for T_3, T_4, T_5 and T_6 are similar. As T_7 is negligible in comparison to the total time complexity, we have $T = \sum_{i=1}^{7} T_i \approx 2^{308.97}$ ($2^{318.81}$) field operations, equivalent to about $2^{295.97}$ ($2^{305.81}$) primitive calls.

4 Design Guidance: Weak Parameters and Secure Patches for AIM2

In the previous section, our cryptanalysis has identified some potential attack vectors for AIM2 construction, *e.g.*, resultant-minimized modeling and subfield reduction, which obviously could give more accurate security bounds of AIM2-III/V against algebraic attacks when compared to the designers' evaluations about Gröbner basis attacks. Especially for AIM2-III, our attack complexity could even touch the claimed security bounds 2^{192} with $2^{188.76}$ when $\omega = 2$ from designers' view.

However, AIM2 already strengthens AIM by adding distinct constants and enlarging the exponents, for which its designers believed would thwart prior successful attacks on AIM [27,30]. Unfortunately, when proposing AIM2 to address these vulnerabilities, the designers did not provide further insights or concrete guidelines on parameter selection, particularly regarding the exponents of the S-boxes. In practice, the designers [22] simply opted for larger exponents (required to be a Mersenne prime) for the Mersenne S-boxes to hinder the construction of low-degree systems of equations, as stated in [22], "increasing the exponents e in Mer[e] substantially enlarges the degree of a system with a moderate number of variables". Based on our analysis, to mitigate resultant-based algebraic attacks and the subfield reduction method, the choice of exponents e_i ($1 \leq i \leq \ell$) in AIM2 should explicitly avoid the following conditions:

- All e_i values are not high enough, which could lower the degree of the resulting system, *e.g.*, even though the exponents in AIM2-III were raised to $e_1 = 17$ and $e_2 = 47$, its security bound against algebraic attacks is lower than the designers expected.
- Some large e_i values being close to the subfield extension degree n/R of $\mathbb{F}_{2^{n/R}}$, which enables effective subfield reduction.

To provide some design guidance of AIM2 construction, in Section 4.1, we identify some weak choices of exponents for AIM2. Then, in Section 4.2, we provide a patched version of AIM2, dubbed AIM2-patch. Related security analysis in Section 4.3 shows that AIM2-patch prevents the use of the resultant-minimized modeling approach and renders the decomposition of linearized polynomials ineffective. Later, performance results in Section 4.4 confirm that this simple patch of AIM2-patch introduces negligible overhead.

4.1 AIM2-weak with Potential Weak Parameters

We evaluate the security of AIM2 under different parameter choices. As outlined in Algorithm 2, we construct equation systems of AIM2 instances with the given parameters via resultant-minimized modeling and solve them using either direct elimination or subfield-based reduction. The resulting complexities identified several weak parameters, summarized in Table 3, which, although specific to our attacks in this paper, can still help in selecting secure parameters and thus provide useful guidance for the design of AIM-like construction.

Interestingly, for the weak instance AIM2-weak-III in Table 3 with $e_1 = 97, e_2 = 101, e_3 = 5$ (larger than the original AIM2-III with $e_1 = 17, e_2 = 47, e_3 = 5$), its security bound degrades from 2^{192} to $2^{168.39}$ when $\omega = 2.373$. This obvious degradation stems from the subfield reduction method, as both Mersenne primes $e_1 = 97$ and $e_2 = 101$ are close to the subfield extension degree $192/2 = 96$.

Algorithm 2 Resultant-Minimized Equation Solving for AIM2

Input: $n, e_1, \ldots, e_\ell$;
Output: $v_1, \ldots, v_\ell$;
Sort $e_1, \ldots, e_\ell$ such that $e_1 > e_2 > \cdots > e_\ell$;
while R is a factor of n **do**
  if $e_1 < n/R$ **then**
    v_1 as the base variable;
    $g_i \leftarrow$ the substitution polynomial of v_i, where $2 \leq i \leq \ell$;
    $f \leftarrow$ the linearized polynomial after substitution;
    return $(v_1, \ldots, v_\ell) \leftarrow$ Algorithm 3 $(g_2, \ldots, g_\ell, f)$.
  else
    $s \leftarrow$ the index satisfying $e_s \geq n/R > e_{s+1}$;
    for $1 \leq i \leq s$ **do**
      $e_i \leftarrow e_i \bmod (n/R)$;
    end for
    v_{s+1} as the base variable;
    $g_i \leftarrow$ the substitution polynomial of v_i, where $i \neq s+1$;
    $f \leftarrow$ the special linearized polynomial;
    $g_{i,1}, \ldots, g_{i,R} \leftarrow$ decompose g_i over $\mathbb{F}_{2^{n/R}}$, where $i \neq s+1$;
    $f_1, \ldots, f_R \leftarrow$ decompose f over $\mathbb{F}_{2^{n/R}}$;
    return $(v_1, \ldots, v_\ell) \leftarrow$ Algorithm 4 $(g_{1,1}, \ldots, f_R)$.
  end if
end while
return $v_1, \ldots, v_\ell$.

4.2 AIM2-patch with a Simple Secure Patch

We now propose a simple patch for AIM2, denoted AIM2-patch, by inserting an additional linear layer L_k after the secret key k. Its 256-bit variant is illustrated in Figure 5. The linearized polynomial L_k is derived from an extendable-output

Algorithm 3 Resultant Elimination

Input: $g_2, \ldots, g_\ell, f$;
Output: $v_1, \ldots, v_\ell$;
$I \leftarrow \langle g_2, \ldots, g_\ell, f \rangle$;
Define the elimination order as $v_\ell \succ \cdots \succ v_2 \succ v_1$;
for $2 \leq i \leq \ell$ **do**
  $I_i = I \cap \mathbb{F}_{2^n}[v_\ell, \ldots, v_{i+1}]$; $\triangleright\ \mathcal{O}((2^{e_i}-1)^\omega \mathcal{M}(d_i))$
end for
Solve the univariate polynomial in v_1; $\triangleright\ \mathcal{U}(d_\ell)$
return $v_1, \ldots, v_\ell$.

Algorithm 4 Subfield-Based Reduction Technique

Input: $g_{1,1}, \ldots, g_{1,R}, \ldots, f_1, \ldots, f_R$;
Output: $v_1, \ldots, v_\ell$;
$I \leftarrow \langle g_{i,1}, \ldots, g_{i,R}, \ldots, f_1, \ldots, f_R \rangle$, where $i \neq s+1$.
Define the elimination order as $v_{1,1} \succ \cdots \succ v_{1,R} \succ \cdots \succ v_{\ell,1} \cdots \succ v_{\ell,R} \succ v_{s+1,1} \succ \cdots \succ v_{s+1,R}$;
for $i = 1, \ldots, s, s+2, \ldots, \ell, s+1$ **do**
  for $j = 1, \ldots, R$ **do**
    $I_{i,j} = I \cap \mathbb{F}_{2^n}[v_{s+1,R}, \ldots, v_{s+1,1}, v_{\ell,R}, \ldots, v_{i,R}, \ldots, v_{i,j+1}]$;
  end for
end for
Solve the univariate polynomial in $v_{s+1,R}$;
Recover $(v_1, \ldots, v_\ell) \leftarrow (v_{1,1}, \ldots, v_{\ell,R})$;
return $v_1, \ldots, v_\ell$.

Table 3: Parameters and attack time complexities (compared to attack models in [28]) of AIM2-weak.

Primitives	e_1	e_2	e_3	e_4	T ($\omega = 2$)	T ($\omega = 2.373$)	Ref.
AIM2-weak-I	67	67	3	-	$2^{109.75}$	$2^{110.79}$	Section 3
	67	67	3	-	$2^{309.98}$	$2^{334.37}$	[28]
AIM2-weak-III	67	131	5	-	$2^{173.65}$	$2^{174.70}$	Section 3
	67	131	5	-	$2^{456.24}$	$2^{500.73}$	[28]
	97	101	5	-	$2^{166.56}$	$2^{168.39}$	Section 3
	97	101	5	-	$2^{460.78}$	$2^{496.67}$	[28]
AIM2-weak-V	137	131	5	3	$2^{243.26}$	$2^{244.38}$	Section 3
	137	131	5	3	$2^{619.52}$	$2^{667.99}$	[28]

† The attack complexity over $\mathbb{F}_2$ is independent of the value of e, and therefore remains unchanged.

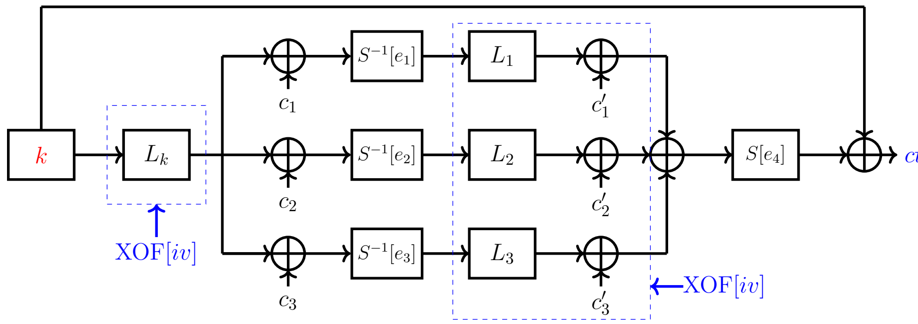

Fig. 5: AIM2-patch-V one-way function.

function (XOF) with initial vector iv, in the same way as L_i ($1 \leq i \leq \ell$). As will be explained in Section 4.3, this simple modification effectively prevents our algebraic attacks. It should be noted that AIM2-patch preserves all other parameters and irreducible polynomials of the original AIM2. The required number of operations and the exhaustive search cost are summarized in Table 4, showing that the additional layer introduces negligible overhead, this can be further confirmed by our performance evaluation in Section 4.4.

Table 4: Number of operations in AIM2 and AIM2-patch, and total exhaustive-search cost (log of binary gates).

Cipher	# Operations		†	Total Cost
	FF Mult.	FF Square	Mat-Vec Mult.	
AIM2-I	14	220	2	144.2
AIM2-patch-I	14	220	3	144.5
AIM2-III	15	242	2	209.1
AIM2-patch-III	15	242	3	209.4
AIM2-V	23	516	3	274.2
AIM2-patch-V	23	516	4	274.5

† FF denotes finite field. The complexity of FF multiplication via FFT is estimated as $n \log n$ binary gates; FF squaring costs n gates; a matrix–vector product costs n^2 gates.

4.3 Algebraic Cryptanalysis on AIM2-patch

Since the number of equations and variables over $\mathbb{F}_2$ remains unchanged, the security of AIM2-patch against algebraic attacks over $\mathbb{F}_2$ is consistent with that of the original AIM2 [22]. In the following, we focus on security analysis of AIM2-patch by considering our attack methods in Section 3.

Analysis for $n = 192$ over $\mathbb{F}_{2^n}$. With notation from Section 3.1, define $L_{b_i} = L_{b_3} \circ L_i$ ($i \in \{1, 2\}$). Introducing variables v_i, v_i' and applying MITM modeling, we obtain the system as below

$$\begin{cases} v_1^{2^{e_1}-1} = L_k(k) + c_1, \\ v_2^{2^{e_2}-1} = L_k(k) + c_2, \\ (v_1' + v_2')^{2^{e_3}-1} = k + ct, \\ L_{b_1}(v_1) + L_{b_3}(v_1') + L_{b_3}(c_1') = 0, \\ L_{b_2}(v_2) + L_{b_3}(v_2') + L_{b_3}(c_2') = 0. \end{cases} \tag{19}$$

Letting $v_3 = v_1' + v_2'$ and $c_3 = c_1' + c_2'$, the above reduces to

$$\begin{cases} g_1 := v_1^{2^{e_1}-1} + L_k(k) + c_1 = 0, \\ g_2 := v_2^{2^{e_2}-1} + L_k(k) + c_2 = 0, \\ g_3 := v_3^{2^{e_3}-1} + k + ct = 0, \\ f := L_{b_1}(v_1) + L_{b_2}(v_2) + L_{b_3}(v_3) + L_{b_3}(c_3) = 0. \end{cases} \tag{20}$$

Then, we let $\deg(L_{b_i}) = 2^{t_i}$ ($1 \le i \le 3$), and require $t_1 + t_2 + t_3 > 2n - 3$ to ensure the existence of a nontrivial solution as discussed in Section 3.1. Unlike in the original AIM2, k cannot be directly eliminated from Equation (20), so the resultant-minimized modeling does not apply. Instead, three successive resultant eliminations are needed to remove v_1, v_2, v_3, yielding a univariate polynomial in k. Although substitution here helps control degrees, the degree of L_k (expected around 2^{n-1}) causes the degree in k to grow excessively, nullifying the advantages of decomposition and substitution. The similar results hold for $n = 128$ and $n = 256$ as given in Table 1, from which one can observe that such a simple patch of AIM2-patch obviously strengthens its resistance against our algebraic attacks.

4.4 Performance Evaluations of AIM2-patch

We now provide the performance evaluations of AIM2-patch under AIMer signature scheme. Note that the transcript format, proof contents, and signature length of AIMer remain unchanged, and the byte length of the public key is also unaffected by the additional linear layer. We benchmarked AIMer with AIM2-patch across three parameter sets (`128f`, `192f`, `256f`) over 10,000 runs. As given in Table 5, compared to the original AIM2 [22], the median signature overhead increased by 2.82%, 1.91%, and 1.49%, while the average overhead rose by 3.14%, 2.23%, and 0.42%, respectively. Overall, the additional layer introduces only a modest overhead of about 1%–3%, which is acceptable in practice.

5 Conclusion

In this paper, we conduct a security analysis of AIM2 over $\mathbb{F}_{2^n}$. We introduce a resultant-minimized modeling, employing a non-k based substitution strategy together with linearized-polynomial decomposition, and combine it with

Table 5: Performance of AIMer with AIM2 vs. AIM2-patch (cycles). Experiments were conducted on a Dell OptiPlex 5090, Intel Core i7-11700 @ 4.80 GHz, 16 GB RAM, Ubuntu 20.04.6, GCC 9.4.0 with `-O3`.

Param.	Median			Average		
	AIM2	AIM2-patch	Δ	AIM2	AIM2-patch	Δ
128f	1,971,100	2,026,598	+**2.82%**	1,994,374	2,056,923	+**3.14%**
192f	5,133,109	5,231,144	+**1.91%**	5,228,079	5,345,064	+**2.23%**
256f	10,792,778	10,953,980	+**1.49%**	11,198,549	11,245,626	+**0.42%**

a subfield-based reduction technique that decomposes high-degree polynomials into lower-degree ones over subfields, thereby greatly simplifying the equation system and reduce the complexity. Specially, the attack time complexity of AIM2-III can be reduced to $2^{188.76}$ primitive calls when $\omega = 2$, while yielding a saving of about 2^{100} primitive calls for AIM2-V. Furthermore, we have identified some weak parameter choices, which could provide concrete design guidance for AIM2 construction, especially for the exponent of its Mersenne S-box.

Our proposed attack method was also applied to AIM2-I, but the results were not satisfactory. This issue will be further investigated in future work. Our attack provides practical guidance for AO primitive design, particularly in parameter selection, and future research will explore its integration with other cryptanalytic techniques to strengthen algebraic constructions.

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A Decomposition of the First Three Equations in Equation (12) over $\mathbb{F}_{2^{128}}$

As an example, consider the decomposition of the first equation:

$$(v_{2,1}+\alpha v_{2,2})^{2^{e_2}} = (v_{2,1}+\alpha v_{2,2})\cdot((v_{1,1}+\alpha v_{1,2})^{2^{e_1}-1}+c_{1,1}+c_{2,1}+\alpha(c_{1,2}+c_{2,2})).$$

In $(v_{1,1}+\alpha v_{1,2})^{2^{e_1}-1}$, both $v_{1,1}$ and $v_{1,2}$ have degrees at most $2^{e_1}-1$, which is derived from

$$\begin{aligned}(v_{1,1}+\alpha v_{1,2})^{2^{e_1}-1} &= (v_{1,1}+\alpha v_{1,2})^{2^1+2^2+\cdots+2^{e_1-1}} \\ &= (v_{1,1}+\alpha v_{1,2})^{2^1}\cdot(v_{1,1}+\alpha v_{1,2})^{2^2}\cdots(v_{1,1}+\alpha v_{1,2})^{2^{e_1-1}} \\ &= (v_{1,1}^2+\alpha^2 v_{1,2}^2)\cdot(v_{1,1}^{2^2}+\alpha^{2^2}v_{1,2}^{2^2})\cdots(v_{1,1}^{2^{e_1-1}}+\alpha^{2^{e_1-1}}v_{1,2}^{2^{e_1-1}}).\end{aligned}$$

Let $\alpha^{2^i}=\alpha+C_i$, the right side of the first equation is then expanded over the subfield as:

$$\begin{aligned}\text{Right} &= (v_{2,1}+\alpha v_{2,2})\cdot(\prod_{i=1}^{e_1-1}(v_{1,1}^{2^i}+\alpha^{2^i}v_{1,2}^{2^i})+c_{1,1}+c_{2,1}+\alpha(c_{1,2}+c_{2,2})) \\ &= (v_{2,1}+\alpha v_{2,2})\cdot(\prod_{i=1}^{e_1-1}(v_{1,1}^{2^i}+(\alpha+C_i)v_{1,2}^{2^i})+c_{1,1}+c_{2,1}+\alpha(c_{1,2}+c_{2,2})) \\ &= v_{2,1}h_{2,1}(v_{1,1},v_{1,2})+v_{2,2}h_{2,2}(v_{1,1},v_{1,2}) \\ &\quad +\alpha(v_{2,1}f_{2,3}(v_{1,1},v_{1,2})+v_{2,2}f_{2,4}(v_{1,1},v_{1,2})),\end{aligned}$$

where the degrees of $v_{1,1}$ and $v_{1,2}$ in $h_{2,1}$, $h_{2,2}$, $f_{2,3}$ and $f_{2,4}$ are at most $2^{e_1}-1$.

Therefore, the first equation can be expanded into the following form:

$$\begin{aligned} v_{2,1}^{2^{e_2}} + C_{e_2} v_{2,2}^{2^{e_2}} + \alpha v_{2,2}^{2^{e_2}} &= v_{2,1} h_{2,1}(v_{1,1}, v_{1,2}) + v_{2,2} h_{2,2}(v_{1,1}, v_{1,2}) \\ &\quad + \alpha (v_{2,1} f_{2,3}(v_{1,1}, v_{1,2}) + v_{2,2} f_{2,4}(v_{1,1}, v_{1,2})), \end{aligned}$$

which can be rewritten as the following system of two equations over $\mathbb{F}_{2^{128}}$:

$$\begin{cases} v_{2,1}^{2^{e_2}} + C_{e_2} v_{2,2}^{2^{e_2}} = v_{2,1} h_{2,1}(v_{1,1}, v_{1,2}) + v_{2,2} h_{2,2}(v_{1,1}, v_{1,2}) \\ v_{2,2}^{2^{e_2}} = v_{2,1} f_{2,3}(v_{1,1}, v_{1,2}) + v_{2,2} f_{2,4}(v_{1,1}, v_{1,2}). \end{cases}$$

By substituting the second equation into the first, we obtain:

$$\begin{cases} v_{2,1}^{2^{e_2}} = v_{2,1} f_{2,1}(v_{1,1}, v_{1,2}) + v_{2,2} f_{2,2}(v_{1,1}, v_{1,2}) \\ v_{2,2}^{2^{e_2}} = v_{2,1} f_{2,3}(v_{1,1}, v_{1,2}) + v_{2,2} f_{2,4}(v_{1,1}, v_{1,2}), \end{cases}$$

where the degrees of $v_{1,1}$ and $v_{1,2}$ in $f_{2,1}$, $f_{2,2}$, $f_{2,3}$ and $f_{2,4}$ are at most $2^{e_1} - 1$.

The decomposition process for the remaining equations is similar. Thus, we obtain

$$\begin{cases} g_{i,1} := v_{i,1}^{2^{e_i}} + v_{i,1} f_{i,1}(v_{1,1}, v_{1,2}) + v_{i,2} f_{i,2}(v_{1,1}, v_{1,2}) = 0, \\ g_{i,2} := v_{i,2}^{2^{e_i}} + v_{i,1} f_{i,3}(v_{1,1}, v_{1,2}) + v_{i,2} f_{i,4}(v_{1,1}, v_{1,2}) = 0, \end{cases}$$

where $\deg_{v_{1,1}}(f_{i,j}) = \deg_{v_{1,2}}(f_{i,j}) \leq 2^{e_1} - 1$ for $2 \leq i \leq 4$ and $1 \leq j \leq 4$.

B Degree Changes and Computational Complexity in Resultant Elimination for Equation (18)

In this section, we further analysis the degrees of $v_{1,1}$ and $v_{1,2}$ during elimination, and derive the associated computational complexity. According to the Algorithm 1, g', F_1', F_2', G_1, G_2, G_3 and G_4 are all updated in each iteration.

Degrees of $v_{1,1}$ and $v_{1,2}$ in Eliminating $v_{2,1}$. In our setting, the size of the Sylvester determinant corresponding to $g' = Res(g_{2,1}, g_{2,2}, v_{2,1})$ is at most $(2^{e_2} + 1) \times (2^{e_2} + 1)$. In this determinant, each entry in the first rows contains $v_{2,2}$ with degree 1 and $v_{1,j}$ ($j \in \{1, 2\}$) with degree $2^{e_1} - 1$, while each entry in the remaining 2^{e_2} rows contains $v_{2,2}$ with degree 2^{e_2} and $v_{1,j}$ with degree $2^{e_1} - 1$. So, we obtain

$$\deg_{v_{2,2}}(g') \leq 2^{2 \cdot e_2} + 1, \ \deg_{v_{1,j}}(g') \leq (2^{e_2} + 1) \cdot (2^{e_1} - 1).$$

The size of the Sylvester determinant corresponding to $G_1 = Res(g_{2,1}, F_1', v_{2,1})$ is at most $3 \cdot 2^{e_2 - 1} \times 3 \cdot 2^{e_2 - 1}$. In this determinant, each entry in the first $2^{e_2 - 1}$ rows contains $v_{2,2}$ with degree 1 and $v_{i,j}$ ($i \in \{3, 4\}$, $j \in \{1, 2\}$) with degree 0, while each entry in the remaining 2^{e_2} rows contains $v_{2,2}$ with degree $2^{e_2 - 1}$ and $v_{i,j}$ with degree $2^{e_i - 1}$. Similarly, each entry in the first $2^{e_2 - 1}$ rows contains $v_{1,1}$

and $v_{1,2}$ with degree $2^{e_1}-1$, while each entry in the remaining 2^{e_2} rows contain $v_{1,1}$ and $v_{1,2}$ with degree D. So, we obtain

$$\deg_{v_{i,j}}(G_1) \leq 2^{e_2+e_i-1},\ \deg_{v_{2,2}}(G_1) \leq 2^{e_2-1} + 2^{2\cdot e_2-1},$$

$$\deg_{v_{1,j}}(G_1) \leq 2^{e_2-1} \cdot (2^{e_1}-1) + 2^{e_2} \cdot D.$$

According to Equation (16), the degree of $v_{1,1}$ and $v_{1,2}$ in the highest-degree cross term $\prod_{i=3}^{4}\prod_{j=1}^{2} v_{i,j}^{2^{e_2+e_i-1}} v_{1,1}^{\deg_{v_{1,1}}(G_1)} v_{1,2}^{\deg_{v_{1,1}}(G_1)}$ of G_1 is at most $4\cdot 2^{e_2+e_1} + \deg_{v_{1,1}}(G_1)$ after substitution. Under our chosen parameters, $4\cdot 2^{e_2+e_1} \ll \deg_{v_{1,1}}(G_1)$, and thus $\deg_{v_{1,1}}(G_1') \approx \deg_{v_{1,1}}(G_1)$. The analysis of resultants $Res(g_{2,1}, F_2', v_{2,1})$ proceeds identically to that of $Res(g_{2,1}, F_1', v_{2,1})$, and thus we omit the details.

The complexity for eliminating $v_{2,1}$ consists of three resultant computations, *i.e.*, computing $Res(g_{2,1}, g_{2,2}, v_{2,1})$, computing $Res(g_{2,1}, F_1', v_{2,1})$ and computing $Res(g_{2,1}, F_2', v_{2,1})$, which can be estimated as

$$T_1 = \mathcal{O}(2 \cdot (3 \cdot 2^{e_2-1})^{\omega} \cdot \mathcal{M}(d_1) + (2^{e_2}+1)^{\omega} \cdot \mathcal{M}(d_2)),$$

where $d_1 = (2^{e_1} - 1 + D)^2 \cdot 2^{e_2-1} \cdot 2^{2e_3-2} \cdot 2^{2e_4-2}$ and $d_2 = 2^{e_2} \cdot (2^{e_1}-1)^2$.

Degrees of $v_{1,1}$ and $v_{1,2}$ in Eliminating $v_{2,2}$. Let us now proceed to the analysis of eliminating $v_{2,2}$. In our setting, the size of the Sylvester determinant corresponding to $G_3 = Res(g', G_1', v_{2,2})$ is at most $(2^{e_2-1} + 3 \cdot 2^{2\cdot e_2-1} + 1) \times (2^{e_2-1} + 3 \cdot 2^{2\cdot e_2-1} + 1)$. In this determinant, each entry in the first $2^{e_2-1} + 2^{2\cdot e_2-1}$ rows contains $v_{i,j}$ $(i \in \{3,4\}\ , j \in \{1,2\})$ with degree 0 and $v_{1,j}$ with degree $\deg_{v_{1,1}}(g')$, while each entry in the remaining $2^{2\cdot e_2}+1$ rows contains $v_{i,j}$ with degree $2^{e_i}-1$ and $v_{1,j}$ with degree $\deg_{v_{1,1}}(G_1')$. So, we obtain

$$\deg_{v_{i,j}}(G_3) \leq (2^{2\cdot e_2}+1) \cdot (2^{e_i}-1),$$

$$\deg_{v_{1,j}}(G_3) \leq (2^{e_2-1} + 2^{2\cdot e_2-1}) \cdot \deg_{v_{1,1}}(g') + (2^{2\cdot e_2}+1) \cdot \deg_{v_{1,1}}(G_1').$$

The degree of $v_{1,1}$ and $v_{1,2}$ remains almost unchanged after substitution. The resultant $Res(g', G_2', v_{2,2})$ is analyzed identically to $Res(g', G_1', v_{2,2})$. Update F_1' and F_2' with G_3' and G_4', respectively.

The complexity for eliminating $v_{2,2}$ consists of two resultant computations, *i.e.*, computing $Res(g', G_1', v_{2,2})$ and $Res(g', G_2', v_{2,2})$, which is estimated as

$$T_2 = \mathcal{O}(2 \cdot (2^{e_2-1} + 3 \cdot 2^{2\cdot e_2-1} + 1)^{\omega} \cdot \mathcal{M}(d_3)),$$

where $d_3 = (\deg_{v_{1,1}}(G_1') + \deg_{v_{1,1}}(g'))^2 \cdot (2^{e_3}-1)^2 \cdot (2^{e_4}-1)^2$.

Degrees of $v_{1,1}$ and $v_{1,2}$ in Eliminating $v_{3,1}$. After eliminating variables $v_{2,1}$ and $v_{2,2}$ and updating polynomials F_1', and F_2', we obtain

$$\begin{aligned}
\deg_{v_{i,j}}(F_1') = \deg_{v_{i,j}}(F_2') &\leq (2^{2\cdot e_2}+1) \cdot (2^{e_i}-1),\\
\deg_{v_{1,j}}(F_1') = \deg_{v_{1,j}}(F_2') &= (2^{e_2-1} + 2^{2\cdot e_2-1}) \cdot (2^{e_2}+1) \cdot (2^{e_1}-1)\\
&\quad + (2^{2\cdot e_2}+1) \cdot (2^{e_2-1} \cdot (2^{e_1}-1) + 2^{e_2} \cdot D),
\end{aligned}$$

where $3 \leq i \leq 4$ and $1 \leq j \leq 2$. In our setting, the size of the Sylvester determinant corresponding to $g' = Res(g_{3,1}, g_{3,2}, v_{3,1})$ is at most $(2^{e_3}+1) \times (2^{e_3}+1)$. In this determinant, each entry in the first rows contains $v_{3,2}$ with degree 1 and $v_{1,j}$ ($j \in \{1,2\}$) with degree $2^{e_1}-1$, while each entry in the remaining 2^{e_3} rows contains $v_{3,2}$ with degree 2^{e_3} and $v_{1,j}$ with degree $2^{e_1}-1$. So, we obtain

$$\deg_{v_{2,2}}(g') \leq 2^{2\cdot e_3}+1,\ \deg_{v_{1,j}}(g') \leq (2^{e_3}+1)\cdot(2^{e_1}-1).$$

The size of the Sylvester determinant corresponding to $G_1 = Res(g_{3,1}, F_1', v_{3,1})$ is at most $(2^{e_3+1}-1) \times (2^{e_3+1}-1)$. In this determinant, each entry in the first $2^{e_3}-1$ rows contains $v_{3,2}$ with degree 1 and $v_{4,j}$ ($j \in \{1,2\}$) with degree 0, while each entry in the remaining 2^{e_3} rows contains $v_{3,2}$ with degree $2^{e_3}-1$ and $v_{4,j}$ with degree $2^{e_4}-1$. Similarly, each entry in the first 2^{e_3-1} rows contains $v_{1,1}$ and $v_{1,2}$ with degree $2^{e_1}-1$, while each entry in the remaining 2^{e_3} rows contains $v_{1,1}$ and $v_{1,2}$ with degree $\deg_{v_{1,1}}(F_1')$. So, we obtain

$$\deg_{v_{4,j}}(G_1) < 2^{e_3+e_4},\ \deg_{v_{3,2}}(G_1) \leq 2^{2e_3}-1,$$

$$\deg_{v_{1,j}}(G_1) \leq (2^{e_3}-1)\cdot(2^{e_1}-1)+2^{e_3}\cdot\deg_{v_{1,1}}(F_1').$$

We apply the substitution

$$\begin{cases} v_{4,1}^{2^{e_4}} = v_{4,1}f_{4,1}(v_{1,1},v_{1,2})+v_{4,2}f_{4,2}(v_{1,1},v_{1,2}) \\ v_{4,2}^{2^{e_4}} = v_{4,1}f_{4,3}(v_{1,1},v_{1,2})+v_{4,2}f_{4,4}(v_{1,1},v_{1,2}) \end{cases}$$

to G_1 and G_2, and denote the substituted polynomials as G_1' and G_2'. According to Equation (16), the degree of $v_{1,1}$ and $v_{1,2}$ in the highest-degree cross term $\prod_{j=1}^{2} v_{4,j}^{2^{e_3+e_4}} v_{1,j}^{\deg_{v_{1,j}}(G_1)}$ of G_1 is at most $2\cdot 2^{e_3+e_1+1}+\deg_{v_{1,1}}(G_1)$ after substitution. Under our chosen parameters, $2\cdot 2^{e_3+e_1+1} \ll \deg_{v_{1,1}}(G_1)$, and thus $\deg_{v_{1,1}}(G_1') \approx \deg_{v_{1,1}}(G_1)$. The analysis of resultants $Res(g_{3,1}, F_2', v_{3,1})$ proceeds identically to that of $Res(g_{3,1}, F_1', v_{3,1})$, and thus we omit the details.

The complexity for eliminating $v_{3,1}$ consists of three resultant computations, *i.e.*, computing $Res(g_{3,1}, g_{3,2}, v_{3,1})$, computing $Res(g_{3,1}, F_1', v_{3,1})$ and computing $Res(g_{3,1}, F_2', v_{3,1})$, which can be estimated as

$$T_3 = \mathcal{O}(2\cdot(2^{e_3+1}-1)^{\omega}\cdot\mathcal{M}(d_1)+(2^{e_3}+1)^{\omega}\cdot\mathcal{M}(d_2)),$$

where $d_1 = (2^{e_1}-1+\deg_{v_{1,1}}(F_1'))^2\cdot(2^{e_3}-1)\cdot(2^{e_4}-1)^2$ and $d_2 = 2^{e_3}\cdot(2^{e_1}-1)^2$.

Degrees of $v_{1,1}$ and $v_{1,2}$ in Eliminating $v_{3,2}$. Let us now proceed to the analysis of eliminating $v_{3,2}$. In our setting, the size of the Sylvester determinant corresponding to $G_3 = Res(g', G_1', v_{3,2})$ is at most $(2^{2e_3+1}) \times (2^{2e_3+1})$. In this determinant, each entry in the first $2^{2e_3}-1$ rows contains $v_{4,j}$ ($j \in \{1,2\}$) with degree 0 and $v_{1,j}$ with degree $\deg_{v_{1,1}}(g')$, while each entry in the remaining $2^{2\cdot e_3}+1$ rows contains $v_{4,j}$ with degree $2^{e_4}-1$ and $v_{1,j}$ with degree $\deg_{v_{1,1}}(G_1')$. So, we obtain

$$\deg_{v_{4,j}}(G_3) \leq (2^{2\cdot e_3}+1)\cdot(2^{e_4}-1),$$

$$\deg_{v_{1,j}}(G_3) = (2^{2e_3} - 1) \cdot \deg_{v_{1,1}}(g') + (2^{2 \cdot e_3} + 1) \cdot \deg_{v_{1,1}}(G_1').$$

The degree of $v_{1,1}$ and $v_{1,2}$ remains almost unchanged after substitution. The resultant $Res(g', G_2', v_{3,2})$ is analyzed identically to $Res(g', G_1', v_{3,2})$. Update F_1' and F_2' with G_3' and G_4', respectively.

The complexity for eliminating $v_{3,2}$ consists of two resultant computations, *i.e.*, computing $Res(g', G_1', v_{3,2})$ and $Res(g', G_2', v_{3,2})$, which is estimated as

$$T_4 = \mathcal{O}(2 \cdot 2^{(2e_3+1) \cdot \omega} \cdot \mathcal{M}(d_3)),$$

where $d_3 = (\deg_{v_{1,1}}(G_1') + \deg_{v_{1,1}}(g'))^2 \cdot (2^{e_4} - 1)^2$.

Degrees of $v_{1,1}$ and $v_{1,2}$ in Eliminating $v_{4,1}$. After eliminating variables $v_{3,1}$ and $v_{3,2}$ and updating polynomials g', F_1', and F_2', we obtain

$$\begin{aligned}
\deg_{v_{4,j}}(F_1') = \deg_{v_{4,j}}(F_2') &\le (2^{2 \cdot e_3} + 1) \cdot (2^{e_4} - 1), \\
\deg_{v_{1,j}}(F_1') = \deg_{v_{1,j}}(F_2') &\le (2^{2e_3} - 1) \cdot \deg_{v_{1,1}}(g') + (2^{2 \cdot e_3} + 1) \cdot \deg_{v_{1,1}}(G_1') \\
&= (2^{2e_3} - 1) \cdot (2^{e_3} + 1) \cdot (2^{e_1} - 1) \\
&+ (2^{2 \cdot e_3} + 1) \cdot ((2^{e_3} - 1) \cdot (2^{e_1} - 1) \\
&+ 2^{e_3} \cdot ((2^{e_2 - 1} + 2^{2 \cdot e_2 - 1}) \cdot (2^{e_2} + 1) \cdot (2^{e_1} - 1) \\
&+ (2^{2 \cdot e_2} + 1) \cdot (2^{e_2 - 1} \cdot (2^{e_1} - 1) + 2^{e_2} \cdot D))),
\end{aligned}$$

where $1 \le j \le 2$. In our setting, the size of the Sylvester determinant corresponding to $g' = Res(g_{4,1}, g_{4,2}, v_{4,1})$ is at most $(2^{e_4} + 1) \times (2^{e_4} + 1)$. In this determinant, each entry in the first rows contains $v_{4,2}$ with degree 1 and $v_{1,j}$ ($j \in \{1, 2\}$) with degree $2^{e_1} - 1$, while each entry in the remaining 2^{e_4} rows contains $v_{4,2}$ with degree 2^{e_4} and $v_{1,j}$ with degree $2^{e_1} - 1$. So, we obtain

$$\deg_{v_{4,2}}(g') \le 2^{2e_4} + 1, \ \deg_{v_{1,j}}(g') \le (2^{e_4} + 1) \cdot (2^{e_1} - 1).$$

The size of the Sylvester determinant corresponding to $G_1 = Res(g_{4,1}, F_1', v_{4,1})$ is at most $(2^{e_4+1} - 1) \times (2^{e_4+1} - 1)$. In this determinant, each entry in the first $2^{e_4} - 1$ rows contains $v_{4,2}$ with degree 1 and $v_{1,j}$ ($j \in \{1, 2\}$) with degree $2^{e_1} - 1$, while each entry in the remaining 2^{e_4} rows contains $v_{4,2}$ with degree $2^{e_4} - 1$ and $v_{1,j}$ with degree $\deg_{v_{1,1}}(F_1')$. So, we obtain

$$\deg_{v_{4,2}}(G_1) \le 2^{2e_4} - 1,$$

$$\deg_{v_{1,j}}(G_1) \le (2^{e_4} - 1) \cdot (2^{e_1} - 1) + 2^{e_4} \cdot \deg_{v_{1,1}}(F_1').$$

The analysis of resultants $Res(g_{4,1}, F_2', v_{4,1})$ proceeds identically to that of $Res(g_{4,1}, F_1', v_{4,1})$, and thus we omit its detailed analysis.

The complexity for eliminating $v_{4,1}$ consists of three resultant computations, *i.e.*, computing $Res(g_{4,1}, g_{4,2}, v_{4,1})$, computing $Res(g_{4,1}, F_1', v_{4,1})$ and computing $Res(g_{4,1}, F_2', v_{4,1})$, which can be estimated as

$$T_5 = \mathcal{O}(2 \cdot (2^{e_4+1} - 1)^{\omega} \cdot \mathcal{M}(d_1) + (2^{e_4} + 1)^{\omega} \cdot \mathcal{M}(d_2)),$$

where $d_1 = (2^{e_1} - 1 + \deg_{v_{1,1}}(F_1'))^2 \cdot (2^{e_4} - 1)$ and $d_2 = 2^{e_4} \cdot (2^{e_1} - 1)^2$.

Degrees of $v_{1,1}$ and $v_{1,2}$ in Eliminating $v_{4,2}$. Let us now proceed to the analysis of eliminating $v_{4,2}$. In our setting, the size of the Sylvester determinant corresponding to $G_3 = Res(g', G_1, v_{4,2})$ is at most $2^{2e_4+1} \times 2^{2e_4+1}$. In this determinant, each entry in the first $2^{2e_4} - 1$ rows contains $v_{1,1}$ and $v_{1,2}$ with degree $\deg_{v_{1,1}}(g')$, while each entry in the remaining $2^{2e_4} + 1$ rows contains $v_{1,1}$ and $v_{1,2}$ with degree $\deg_{v_{1,1}}(G_1)$. So, we obtain

$$\deg_{v_{1,1}}(G_3) = (2^{2e_4} - 1) \cdot \deg_{v_{1,1}}(g') + (2^{2\cdot e_4} + 1) \cdot \deg_{v_{1,1}}(G_1).$$

The resultant $Res(g', G_2, v_{2,2})$ is analyzed identically to $Res(g', G_1, v_{2,2})$. Update F'_1 and F'_2 with G'_3 and G'_4, respectively.

The complexity for eliminating $v_{2,2}$ consists of two resultant computations, *i.e.*, computing $Res(g', G'_1, v_{4,2})$ and $Res(g', G'_2, v_{4,2})$, which is estimated as

$$T_6 = \mathcal{O}(2 \cdot 2^{(2e_4+1)\cdot\omega} \cdot \mathcal{M}(d_3)),$$

where $d_3 = (\deg_{v_{1,1}}(G'_1) + \deg_{v_{1,1}}(g'))^2$.

C Details of Attacking AIM2-I

Directly transferring the attack method from AIM2-III to AIM2-I causes the degree of the basis variables after substitution to exceed 2^{128} and makes linearized polynomial decomposition infeasible, primarily due to the excessively large parameters e_1 and e_2. In this subsection, the subfield-based reduction method and the squaring modulo reduction method are applied separately to the analysis of AIM2-I for further optimization.

Subfield-based Reduction Method. Since the modeling of AIM2-I is consistent with that of AIM2-III, we construct the following system of equations.

$$\begin{cases} g_1 := v_1^{2^{e_1}} + v_1 \cdot (v_3^{2^{e_3}-1} + c_1 + ct) = 0, \\ g_2 := v_2^{2^{e_2}} + v_2 \cdot (v_3^{2^{e_3}-1} + c_2 + ct) = 0, \\ f := L_{b_1}(v_1) + L_{b_2}(v_2) + L_{b_3}(v_3) + L_{b_3}(c) = 0. \end{cases} \tag{21}$$

To reduce the degree of the equations, we transfer the equation system to the subfield $\mathbb{F}_{2^{16}}$ and derive low-degree representation for Equation (21). Precisely, let

$$L_{b_3}(x) = \sum_{i=0}^{127} a_i x^{2^i}, L_1(x) = \sum_{j=0}^{127} \gamma_{1,j} x^{2^j}, L_2(x) = \sum_{j=0}^{127} \gamma_{2,j} x^{2^j}$$

where $\gamma_{1,j}$'$s, \gamma_{2,j}$'$s \in \mathbb{F}_{2^{128}}$ are known coefficients, while a_i'$s \in \mathbb{F}_{2^{128}}$ are the unknowns to be determined. To accommodate our attack stratety on the subfield $\mathbb{F}_{2^{16}}$, we decompose polynomial L_{b_i} $(1 \leq i \leq 3)$ into eight parts and the intuitive representation of L_{b_3} is shown in Figure 6. Specially, we expect

$$\underbrace{\boxed{a_0 + \cdots + a_{t_3}x^{2^{t_3}}} + 0x^{2^{t_3+1}} + \cdots + \boxed{a_{16}x^{2^{16}} + \cdots + a_{2^{16+t_3}}x^{2^{16+t_3}}} + 0x^{2^{17+t_3}} + \cdots + \boxed{a_{112}x^{2^{112}} + \cdots + a_{112+t_3}x^{2^{112+t_3}}} + \cdots + 0x^{2^{127}}}_{\deg(L_{b_4}) = 2^{112+t_3} \quad \text{over} \quad \mathbb{F}_{2^{128}}}$$

$$\downarrow$$

$$\underbrace{\begin{array}{c} \boxed{a_0x + a_1x^2 + \cdots + a_{t_3}x^{2^{t_3}}} \\ + \\ \boxed{a_{16}x + a_{17}x^2 + \cdots + a_{16+t_3}x^{2^{t_3}}} \\ + \\ \vdots \\ \boxed{a_{112}x + a_{113}x^2 + \cdots + a_{112+t_3}x^{2^{t_3}}} \end{array} \Longrightarrow (a_0 + a_{16} + \cdots + a_{112})x + (a_1 + \cdots + a_{113})x^2 + \cdots + (a_{t_3} + a_{16+t_3} + \cdots + a_{112+t_3})x^{2^{t_3}}}_{\deg(L_{b_4}) = 2^{t_3} \quad \text{over} \quad \mathbb{F}_{2^{16}}}$$

Fig. 6: The intuitive representation of $L_{b_3}(x)$ over $\mathbb{F}_{2^{128}}$ and $\mathbb{F}_{2^{16}}$.

$$\begin{cases} L_{b_3}(x) = \sum_{l=0}^{7} \sum_{i=l\cdot 16}^{t_3+l\cdot 16} a_i x^{2^i}, \\ L_{b_1}(x) = \sum_{l=0}^{7} \sum_{\delta=l\cdot 16}^{l\cdot 16+t_1} \sum_{\substack{0\le i,j\le 127 \\ i+j=\delta \bmod 128}} a_i \gamma_{1,j}^{2^i} x^{2^\delta}, \\ L_{b_2}(x) = \sum_{l=0}^{7} \sum_{\delta=l\cdot 16}^{l\cdot 16+t_2} \sum_{\substack{0\le i,j\le 127 \\ i+j=\delta \bmod 128}} a_i \gamma_{2,j}^{2^i} x^{2^\delta}, \end{cases}$$

where $0 \le t_1, t_2, t_3 \le 16$, $a_{t_3+l\cdot 16}$, $\sum_{\substack{0\le i,j\le 127 \\ i+j=t_{\eta+l\cdot 16} \bmod 128}} a_i \gamma_{\eta,j}^{2^i} \neq 0$ $(1 \le \eta \le 2,\ 0 \le l \le 7)$. The constraint $t_1 + t_2 + t_3 + t_4 > 2 \cdot 16 - 3$ must hold to ensure the existence of the polynomial L_{b_i} $(1 \le i \le 3)$. Then, Equation Equation (21) over $\mathbb{F}_{2^{128}}$ constitutes a system of 3 equations in 3 variables, including one equation of degree 2^{49}, one of degree 2^{91} and one of degree $\max(2^{112+t_1}, 2^{112+t_2}, 2^{112+t_3})$.

After reducing e_1 from 49 to 1 and e_2 from 91 to 11, Equation (21) can be decomposed to 48 equations in 48 variables over $\mathbb{F}_{2^{16}}$ as

- 16 equations of degree 2^1;
- 16 equations of degree 2^{11};
- 16 equations of degree $\max(2^{t_1}, 2^{t_2}, 2^{t_3})$.

Due to the excessive number of variables and equations, the resultant elimination method yields unsatisfactory results.

Squaring Modulo Reduction Method. We rewritten Equation (21) as

$$\begin{cases} g_1 := v_2^{2^{e_2}} + v_2 \cdot (v_1^{2^{e_1}-1} + c_1 + c_2) = 0, \\ g_2 := v_3^{2^{e_3}-1} + v_1^{2^{e_1}-1} + c_1 + ct = 0, \\ f := L_{b_1}(v_1) + L_{b_2}(v_2) + L_{b_3}(v_3) + L_{b_3}(c) = 0, \end{cases}$$

then perform $128 - e_2$ times the square operation on the first equation and yield

$$v_2 = (v_1^{2^{e_1}-1} + c_1 + c_2)^{2^{128-e_2}} \cdot v_2^{2^{128-e_2}}.$$

Consequently, we reduce the degree of v_2 in the first equation from 2^{91} to 2^{37} and decrease the size of the determinant during its elimination, thereby further reducing the time complexity of resultant computation.

Let $\deg(L_{b_i}) = 2^{t_i}$ $(1 \le i \le 3)$ and the condition $t_1 + t_2 + t_3 > 2 \cdot 128 - 3$ should be held to ensure the existence as in Section Section 3.1. According to Equation Equation (8), we employ the substitution $v_3^{2^{e_3}} = v_3(v_1^{2^{e_1}-1} + c_1 + ct)$ to $L_{b_3}(v_3)$ and denote the substituted polynomial as $L'_{b_3}(v_3, v_1)$, where $\deg_{v_1}(L'_{b_3}) < 2^{t_3-e_3+e_1+1}$ and $\deg_{v_3}(L'_{b_3}) \le 2^{e_3-1}$. Let

$$\begin{cases} g'_1 := v_2 + (v_1^{2^{e_1}-1} + c_1 + c_2)^{2^{n-e_2}} \cdot v_2^{2^{n-e_2}} = 0, \\ g_2 := v_3^{2^{e_3}-1} + v_1^{2^{e_1}-1} + c_1 + ct = 0, \\ f' := L_{b_1}(v_1) + L_{b_2}(v_2) + L'_{b_3}(v_3, v_1) + L_{b_3}(c) = 0, \end{cases} \tag{22}$$

where $\deg_{v_1}(f') \le \max(2^{t_1}, 2^{t_3-e_3+e_1+1})$. The details of eliminating variables and deriving the univariate polynomial in v_1 is outlined as follows:

Step 1. First, we calculate $G_1(v_1, v_3) = Res(g'_1, f', v_2)$ to eliminate v_2. The size of the Sylvester determinant corresponding to $G_1(v_1, v_3)$ is at most $(2^{n-e_2} + 2^{t_2}) \times (2^{n-e_2} + 2^{t_2})$. In this determinant, each entry in the first 2^{t_2} rows contains the variable v_3 with degree 0 and v_1 with degree $(2^{e_1}-1) \cdot 2^{n-e_2}$, while each entry in the remaining 2^{n-e_2} rows contains v_3 with degree 2^{e_3-1} and v_1 with degree $\deg_{v_1}(f')$. So, we have

$$\deg_{v_3}(G_1) \le 2^{n-e_2} \cdot 2^{e_3-1},$$

$$\deg_{v_1}(G_1) \le 2^{t_2} \cdot (2^{e_1} - 1) \cdot 2^{n-e_2} + 2^{n-e_2} \cdot \deg_{v_1}(f').$$

We employ the substitution $v_3^{2^{e_3}} = (v_1^{2^{e_1}-1} + c_1 + ct) \cdot v_3$ to $G_1(v_1, v_3)$ and denote the substituted $G_1(v_1, v_3)$ as $G'_1(v_1, v_3)$, it becomes evident that $\deg_{v_3} \le 2^{e_3} - 2$. According to Equation Equation (8), the degree of v_1 in the highest-degree cross term $v_1^{\deg_{v_1}(G_1)} v_3^{\deg_{v_3}(G_1)}$ of G_1 is at most $\deg_{v_3}(G'_1) \le 2^{n-e_2+e_1} + \deg_{v_1}(G_1)$ after substitution.

Step 2. Secondly, we calculate $G_2(v_1) = Res(g_2, G'_1, v_3)$ to eliminate v_3. The size of the Sylvester determinant corresponding to $G_2(v_1)$ is at most $(2^{e_3+1} - 3) \times (2^{e_3+1} - 3)$. In this determinant, each entry in the first $2^{e_3} - 2$ rows contains the variable v_1 with degree $2^{e_1} - 1$, while each entry in the remaining $2^{e_3} - 1$ rows contains v_3 with degree at most $\deg_{v_1}(G'_1)$. So, we have

$$\deg_{v_1}(G_2) \le (2^{e_3} - 2) \cdot (2^{e_1} - 1) + (2^{e_3} - 1) \cdot \deg_{v_1}(G'_1).$$

Step 3. Finally, we solve the univariate polynomial with respect to the variable v_1.

Complexity Analysis. The size of $Res(g'_1, f', v_2)$ is at most $(2^{n-e_2} + 2^{t_2}) \times (2^{n-e_2} + 2^{t_2})$ and the complexity can be estimated as

$$T_1 = \mathcal{O}((2^{n-e_2} + 2^{t_2})^{\omega} \cdot \mathcal{M}(d)),$$

where $d = (2^{e_3-1}+1)\cdot((2^{e_1}-1)\cdot(2^{n-e_2})+\deg_{v_1}(f')+1)$.

The size of $Res(g_2, G'_1, v_3)$ can be further reduced to $(2^{e_3}-1)\times(2^{e_3}-1)$ and each entry has degree at most $2^{e_1}-1+\deg_{v_1}(G'_1)$. Consequently, the time complexity in this step can be estimated as

$$T_2 = \mathcal{O}((2^{e_3}-1)^{\omega}\cdot\mathcal{M}(d)),$$

where $d = 2^{e_1}-1+\deg_{v_1}(G'_1)$.

Finally, we compute the complexity of solving the univariate polynomial $G_2(v_1)$, which can be estimated as

$$T_3 = \mathcal{O}(\deg_{v_1}(G_2)\log(\deg_{v_1}(G_2))(\log(\deg_{v_1}(G_2)) + \log 2^n \log\log(\deg_{v_1}(G_2)))).$$

The overall time complexity of our attack can be estimated as $T = T_1 + T_2 + T_3$.

Parameter Selection. The overall complexity is highly dependent on $\deg_{v_1}(f')$. This requires balancing 2^{t_1} and $2^{t_3-e_3+e_1+1}$ as close as possible under the condition $t_1+t_2+t_3+3>2n$. Specially, we choose $t_1 = 126,\ t_2 = 49,\ t_3 = 79$.

Open Problems. Based on the above analysis, we have identified two problems, which are considered the breakthroughs for our subsequent attack on AIM2-I.

- Are there more efficient methods to solve the low-degree equations constructed via the subfield-based reduction method?
- Is it possible to combine the square-degree-reduction method with other techniques to further optimize attacks on AIM2-I?